

WJEC GCE Physics

A2 UNIT 4 PAST PAPER BY TOPIC

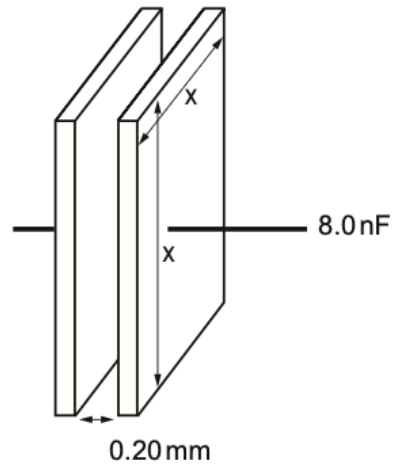
IGC HK Exam



Capacitors

2017

1. (a) (i) An 8.0 nF capacitor has square plates separated by 0.20 mm of air. Calculate the length (x) of the sides of the plates. [3]



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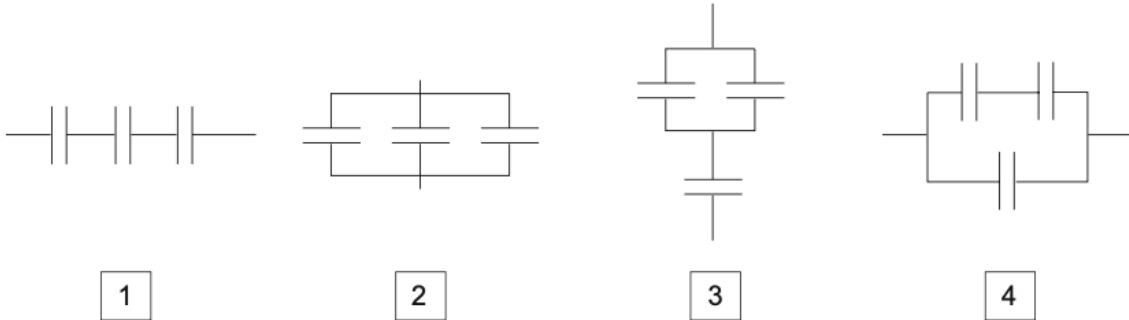
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- (ii) Which of the following combinations of 8.0 nF capacitors has a capacitance of 12.0 nF ? Justify your answer. [4]

All individual capacitors shown are 8.0 nF .



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- (iii) State how the capacitance of the capacitor can be increased without changing its dimensions. [1]

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- (iv) Calculate the charge on the plates of the 8.0 nF capacitor when it stores an energy of $0.62\text{ }\mu\text{J}$. [3]

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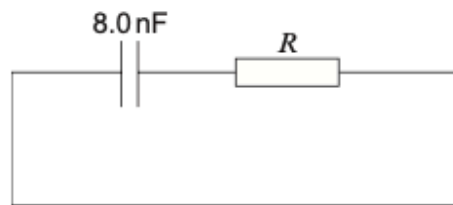
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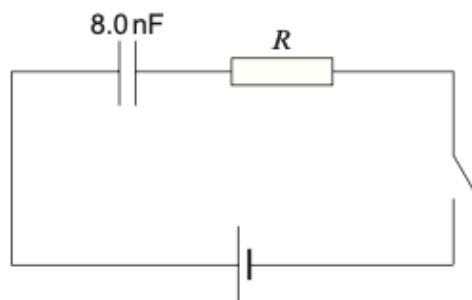
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- (b) (i) A charged 8.0 nF capacitor is discharged through a large resistor. Calculate the resistance of the resistor if the time constant is 15.5 ms . [2]



- (ii) Calculate the fraction of the initial charge remaining on the capacitor after 31.0 ms . [3]

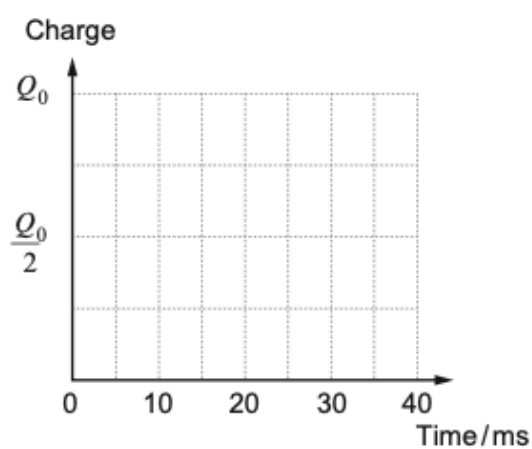
- (c) An identical, uncharged 8.0 nF capacitor is **charged** in the following circuit, **using the same large resistor, R** . The switch is closed and the capacitor is uncharged at time $t = 0$.



Without further calculations, sketch graphs of:

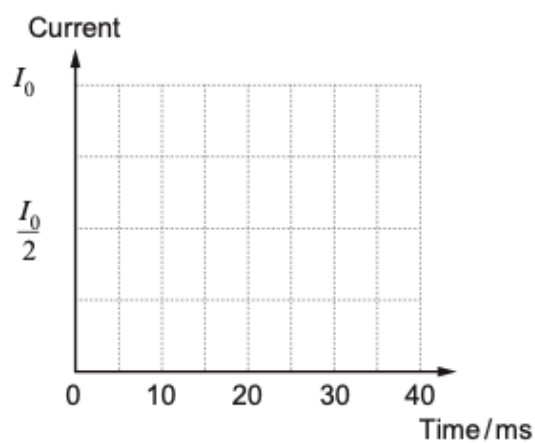
- (i) the charge on the plates of the capacitor against time;

[2]



- (ii) the current in the circuit against time.

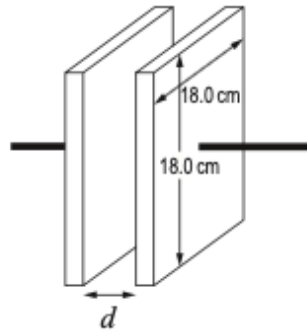
[2]



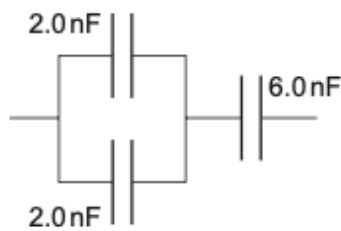
Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
1	(a)	(i)	Capacitor equation rearranged $\frac{Cd}{\epsilon_0} = A$ (1) Area = x^2 used (1) Correct answer = 42.5 [cm] (1)	1	1				
		(ii)	Logical method employed e.g. eliminating 1, 2 etc.(can be implied by correct calculation for 4) (1) Evidence of a correct capacitor in series calculation (1) Evidence of a correct capacitor in parallel calculation (1) Number 4 chosen (1)		1		3	3	
		(iii)	Dielectric / increase the permittivity Accept a named dielectric	1		4	4	2	
		(iv)	Valid method e.g. $\frac{q^2}{2C} = U$ or 2 other valid equations (1) Accept use of 3 equations Rearrangement or algebra (1) Answer = 9.96×10^{-8} [C] (accept 1×10^{-7} C) (1)	1					
	(b)	(i)	RC used as the time constant (1) Answer = 1.94 M[Ω] (1)	1	1		2	2	
		(ii)	2 time constants interpreted or substitution into: $q = q_0 e^{-t/CR}$ (1) 0.37 ² or rearrangement of equation ecf (1) Answer = 0.135 or 13.5% or $\frac{27}{200}$ (1)		3		3	3	
	(c)	(i)	Correct shape (starting from 0,0 increasing with decreasing gradient) (1) 50-75% charged at 15.5 ms or 8-15 ms half-life (1)	1	1		2	2	2
		(ii)	Correct shape (starting from I_0 , 0 decreasing and decreasing negative gradient) (1) 50-75% discharged at 15.5 ms or 8-15 ms half-life (1)	1	1		2	2	2
			Question 1 total	6	10	4	20	17	4

2018

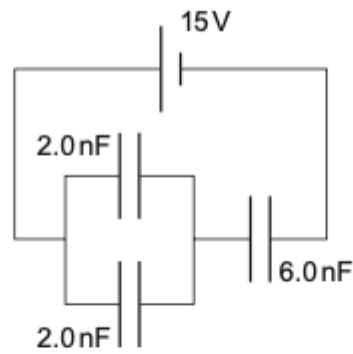
1. (a) (i) A laboratory technician has two square aluminium plates with sides of length 18.0 cm. Calculate the separation, d , of the plates that produces a capacitance of 2.0 nF. [3]



- (ii) Calculate the capacitance of the capacitor combination shown. [3]



- (iii) Show that the charge stored on a 2.0 nF capacitor in the following circuit is 18 nC . [3]



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- (iv) Calculate the total energy stored by the capacitor combination. [2]

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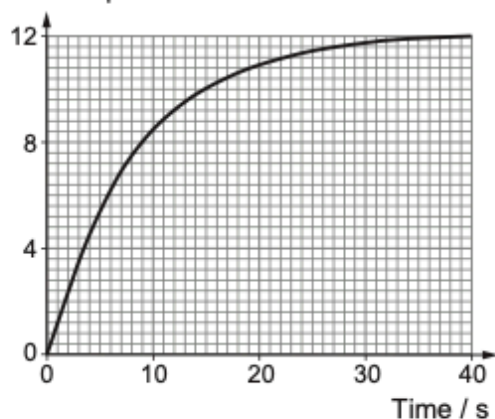
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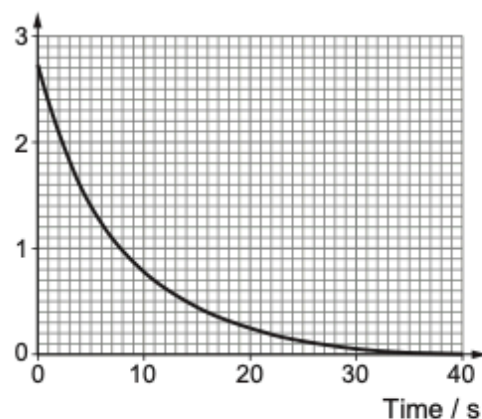
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- (b) A group of students investigated the **charging** of a capacitor through a resistor and obtained the following data.

pd across capacitor / V



Current / mA



- (i) Draw a circuit diagram of the apparatus that might have been used to obtain the data. [2]

- (ii) Use the graphs to determine:
- the emf of the power supply;
 - the resistance of the resistor;
 - the capacitance of the capacitor.

[5]

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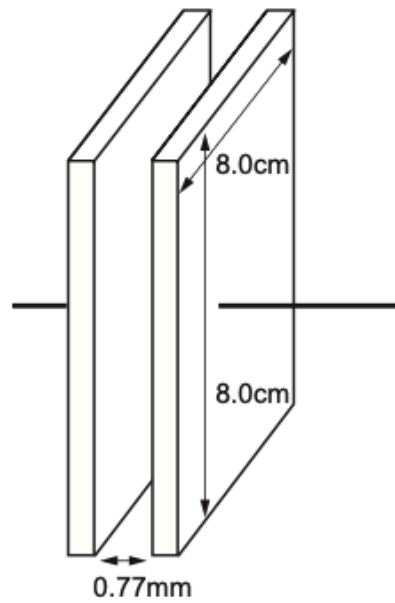
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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
1	(a)	(i)	Capacitor equation rearranged $d = \frac{\epsilon_0 A}{C}$ (1) Substitution of values (1) Correct answer 1.43×10^{-4} m (0.143 mm) (1) [Accept 1.4×10^{-4} m, not 1×10^{-4}]	1	1 1		3	3	
		(ii)	Adding capacitors in parallel (4 nF) (1) Understanding of 6 nF in series with 4 nF i.e. correct application of physics (1) [even failure to invert to get answer, so $1/4 + 1/6 \rightarrow 2$ marks] Correct answer 2.4 n[F] or 12/5 n[F] (1) c.a.o.	1	1 1		3	2	
		(iii)	Any valid CV e.g. 2×9 , 6×6 , 2.4×15 (1) [Accept 2.4×15 or $\frac{1}{2} \times 2.4 \times 15$; not 1.2×15 or 2.4×7.5] $\frac{1}{2}$ -way explanation: charge split / divided between 2 nFs / 9 V across 2 nF / 6 V across 6 nF (1) Completed explanation: all charge on capacitors in series / justify 9 V or 6V [e.g. $V \propto 1/C$]	1	1 1		3	2	
		(iv)	<u>Correct</u> substitution (into valid equation) e.g. $0.5 QV$ etc. (1) Correct answer (270 nJ or 2.7×10^{-7} J) (1)	1	1		2	2	
	(b)	(i)	Capacitor in series with resistor and cell/battery/psu (1) Properly placed ammeter and voltmeter (1)	2			2		2

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
		(ii)	Emf = 12 V stated [accept $V_{\max}$] (1) Initial current = 2.7 mA (1) $R = \frac{V}{I} = 4444 \Omega$ (1) [4.4 k Ω] [N.B. With incompatible V and I, 2nd and 3rd marks not accessible] $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at the origin with $I = 0$ (or $V = 12$ V) OR taking logs and rearranging OR obtaining Q from total area under current graph around 2.0×10^{-2} C (1) $C = \frac{8.0}{4400} = 1.8 [\pm 0.3]$ mF (1) [No ecf] Unit penalty (-1): all 3 units [V, Ω , F] needed Alternative Emf = 12 V (1) Find $\frac{dV}{dt}$ and I at given t , e.g. at 10 s, 0.411 V s $^{-1}$, 7.5 mA (1) Hence calculate C from $I = C \frac{dV}{dt} \rightarrow 1.8$ mF ecf (1) $RC \sim 8$ s from 37% $I_{\max}$ value or 63% $V_{\max}$ value or intercept of gradient at $t = 0$ with $I = 0$ (or $V = 12$ V) (1) R from C and $RC \rightarrow 4.4$ k Ω (1) Unit penalty (-1): all 3 units [V, Ω , F] needed		5		5	5	5
			Question 1 total	6	12	0	18	14	7

2019

1. (a) (i) For the air-spaced parallel plate capacitor shown, calculate the pd applied when it stores $25.5\mu\text{J}$ of energy. [4]



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(ii) Explain why the capacitor stores energy when a pd is applied to the plates. [2]

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(iii) A group of scientists claim that they have developed a new dielectric that enables the above capacitor to store a million times more charge and energy for a given pd. Explain what further steps must be taken by the scientific community and industry before this new dielectric can be used in devices and sold to the public. [3]

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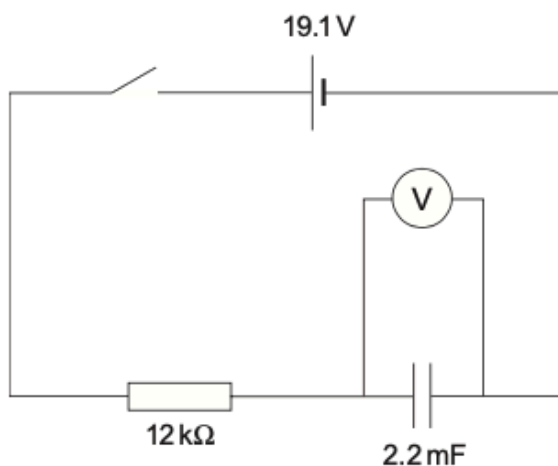
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(b) Bethan investigates the charging of a capacitor using the following circuit.



The results she obtains are then tabulated.

Time / s	pd across capacitor / V $\pm 0.5\text{V}$	Charge on capacitor / mC $\pm 1.1\text{mC}$
10.0	6.1	13.4
20.0	10.0	22.0
30.0	12.8	
40.0	15.1	33.2
50.0	16.2	35.6
60.0	17.1	
70.0	17.7	38.9
80.0	18.3	40.3

- (i) Confirm that the uncertainty in the charge is 1.1 mC (you may assume that the uncertainty in the 2.2mF capacitor is negligible).

[1]

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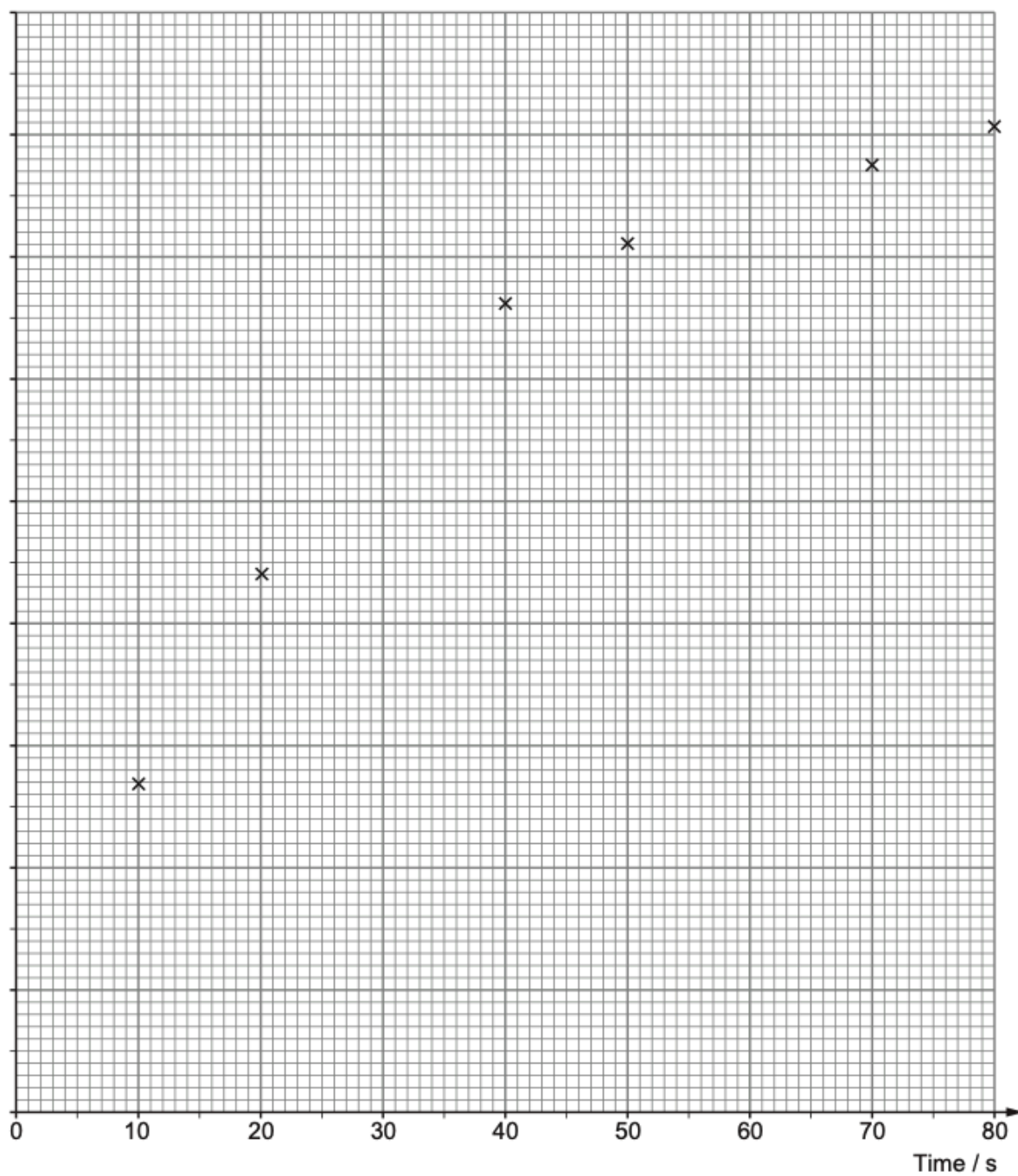
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- (ii) **Complete** the table.

[2]

- (iii) Complete the graph by labelling the y -axis scale, plotting the remaining two points, adding error bars for charge and drawing a curve of best fit. [5]

Charge / mC



- (iv) **Use the curve of best fit** to obtain the time constant of the charging circuit (show your working and do not simply multiply the resistance by the capacitance). [3]

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- (v) **By drawing a suitable tangent**, calculate the current in the circuit at 45s. [2]

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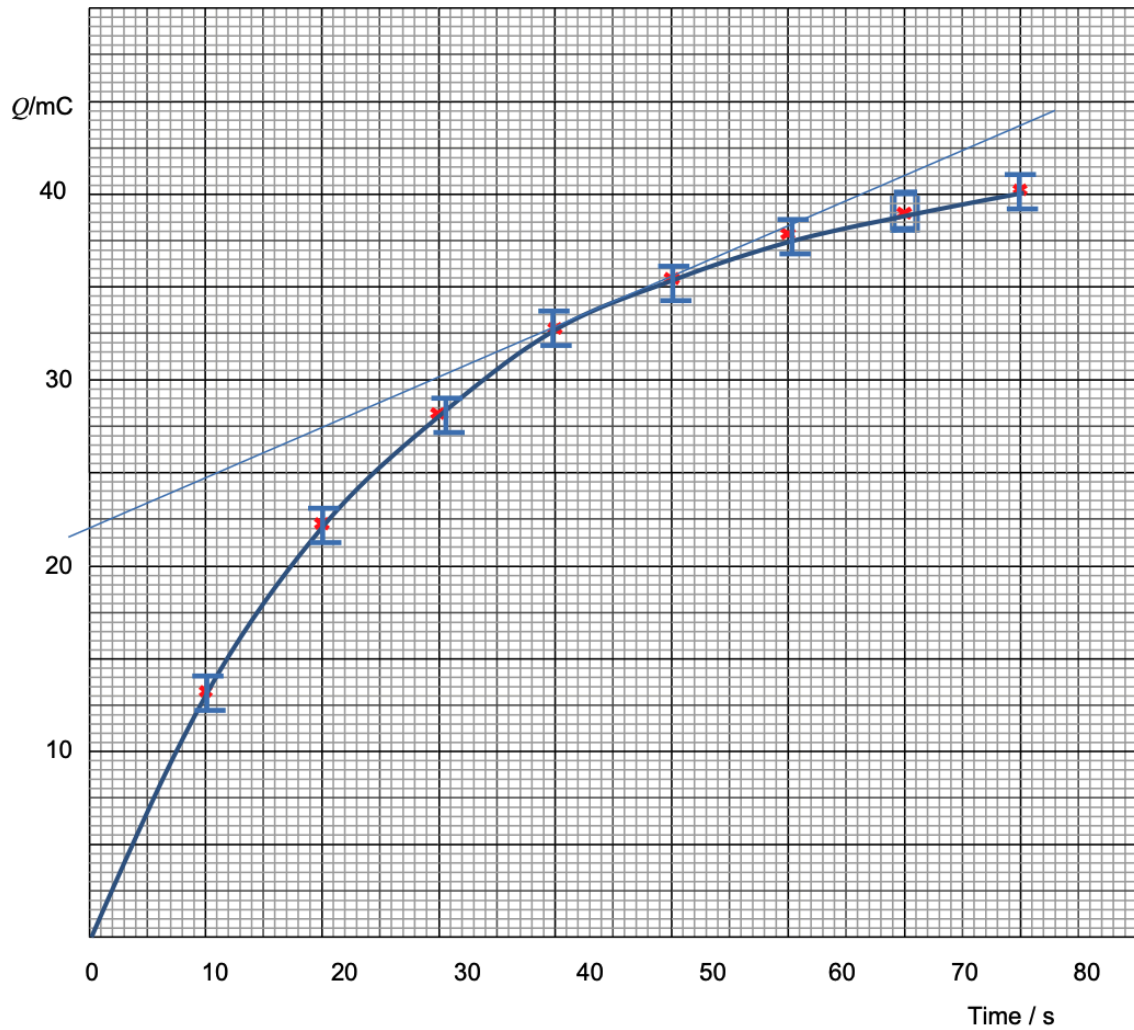
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$$Q = Q_0 \left(1 - e^{-\frac{t}{RC}} \right) \text{ and } I = I_0 e^{-\frac{t}{RC}}$$

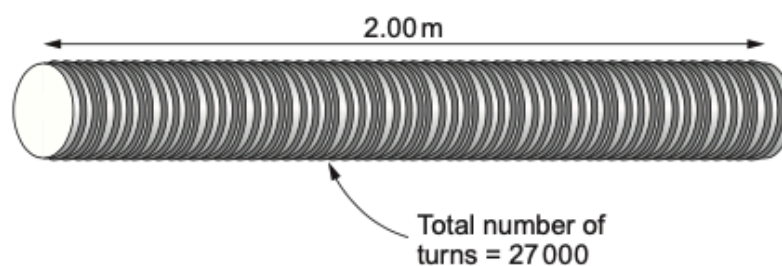
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Use of $C = \frac{\epsilon_0 A}{d}$ [73.6 pF] [1] Use of $E = \frac{1}{2} CV^2$ or combination of $E = \frac{1}{2} QV$ and $Q = CV$ [1] Rearrangement $\left(\frac{2Ed}{\epsilon_0 A} \right)$ OR substitution in both equations [1] Correct answer = 833 [V] [1]	1 1	 1 1		 4	 3	
		(ii)	[Pd supplies/leads to] charges on plates (1) accept electric field set-up [This set-up] can do work / [electrostatic] PE stored / energy stored <u>in field</u> (1) accept can provide current after pd removed	2			2		
		(iii)	Any 3 (x 1) valid points e.g. • Repeat experiment / more tests / repeatability • Repeats by other research groups / industry / reproducibility • Tests for safety [of dielectric] • Tests for lifetime [of dielectric] • Cost effectiveness / availability • Environmental impact • Check for charge leakage			3	3		
	(b)	(i)	2.2 mF x 0.5 V seen or clear % usage shown		1		1		1
		(ii)	28.2 (1) 37.6 (1)		2		2		2
		(iii)	Numbers on y-axis i.e. 10, 20, 30, 40 (1) Both points plotted correctly ecf < small square (1) y-error bars correct < small square (1) Good curve of best fit ecf (2) If flawed but valid attempt at best fit e.g. hairy line, not smooth, missing one of the bars award 1 mark only		5		5		5

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(iv)		Obtaining $Q_0 = 19.1 \times 2.2 \text{ mF} = 42 \pm 2 \text{ m[C]}$ (1) Using 63 % of fully charged OR substituting values into $Q = Q_0 \left(1 - e^{-\frac{t}{RC}}\right)$ OR taking logs correctly (1) T or $RC = 27 \pm 2 \text{ [s]}$ (no marks for 26.4 s) OR Q value correct after substituting (1)		3		3	3	3
	(v)		Good tangent drawn at 45 s ecf (1) Gradient = 0.27 m[A] ecf (1)		2		2		2
	(c)		Any 5 × (1) from: - Shape of graph is good - Line of best fit goes through error bar(s) - $RC = 26.4 \text{ s}$ - RC in good agreement with 27 s (ecf available) - $I_0 = \frac{19.1}{12\,000} = 1.59 \text{ m[A]}$ - Gradient of graph decreases / gradient is the current - So current equation is in good agreement – only award mark if current decreasing implied - Substitution into either equation for confirmation			5	5	2	5
			Question 1 total	4	15	8	27	8	18



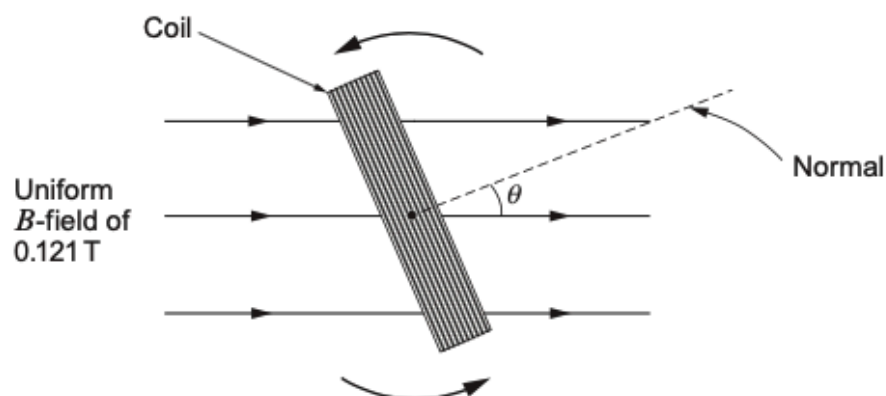
2022

1. (a) (i) Lindsey calculates that a current of 3.57 A will produce a magnetic flux density of 0.121 T inside the long solenoid shown. Determine whether or not she is correct. [3]



- (ii) State how the magnetic flux density inside the solenoid can be increased greatly without changing the current or the number of turns per unit length. [1]

- (b) A rectangular coil rotates at a constant angular velocity within a uniform magnetic field of 0.121 T. The coil has 70 turns and cross-sectional area 59 cm². The diagram below shows the coil, looking along the axis of rotation.



- (i) Calculate the flux linkage of the coil when $\theta = 23^\circ$. [2]

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- (ii) As the coil rotates, explain what values of θ provide the maximum and minimum values of the induced emf. [2]

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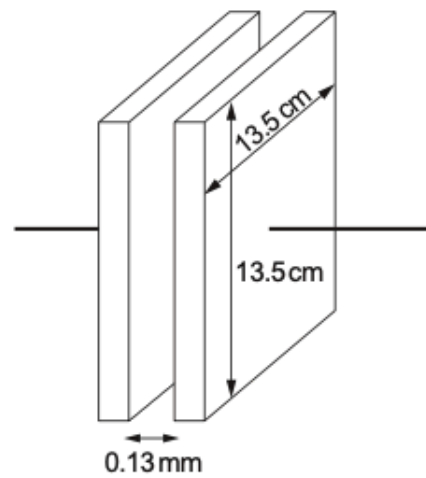
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i)	Valid method e.g. $B = \mu_0 n I$ to calculate B or I or n or μ_0 (1) Gives $B = 0.0606$ [T] or $I = 7.13$ [A] or $n = 26\,972$ or $\mu_0 = 2.51 \times 10^{-6}$ (1) Correct conclusion based on calculation (with ecf) e.g. Lindsay wrong (if B or I calculated), Lindsay wrong because μ_0 too large or real value is 1.26×10^{-6} or Lindsay wrong because n value too large or real value is 13 500 (allow ecf on $n = 27\,000$ here) (1) i.e. Calculating $B = 0.121$ [T] followed by Lindsay correct is 2 marks (1 st and 3 rd marks)			3	3	1	
		(ii)	Iron / ferromagnetic / steel core Accept increase permeability Don't accept increase permeability of free space	1			1		
	(b)	(i)	Use of $BAN\cos\theta$ (1) Correct answer = 0.046 Wb or Wb turn or T m ² **unit mark** (1)	1	1		2	1	
		(ii)	Max when $\theta = 90$ because flux changing / cutting at max rate (accept cutting most lines or similar for max rate) (1) When $\theta = 0$ no emf because flux max or no cutting or rate of change is zero (1) Alternative for 2nd mark Min when $\theta = -90, 270$ because flux changing / cutting in opposite direction (1) Allow 1 mark if both angles are correct with no explanation Allow 1 mark if angles are wrong but both explanations are correct (must refer to flux changing or cutting but no need to insist on "rate")		2		2		
			Question 1 total	2	3	3	8	2	0

4. (a) Calculate the capacitance of the capacitor shown.

[2]



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- (b) (i) Calculate the pd across the terminals of a 5.0 mF capacitor when it stores 1.00 J of energy.

[2]

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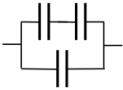
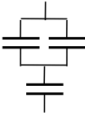
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- (ii) In the space provided, draw a diagram to show how you would combine three 5.0 mF capacitors to produce a capacitance of 7.5 mF. [2]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Substitution into: $C = \frac{\epsilon_0 A}{d}$ (1) Correct answer = 1.24 n[F] (1)	1	1		2	1	
	(b)	(i)	Combining $0.5QV$ and $Q = CV$ (or using $0.5CV^2$) (1) Correct answer = 20 [V] (1)		2		2	2	
		(ii)	2 capacitors in series all in parallel with another  Allow 1 mark for  Subtract a mark if the symbol for capacitor is wrong			2	2	2	

2023

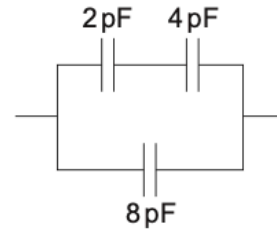
1. (a) Calculate the total capacitance of each of the following capacitor combinations. [6]



(i)



(ii)



(iii)

(i)

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(ii)

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(iii)

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- (b) Michael, a laboratory technician, must make a $2.7 \mu\text{F}$ capacitor using only two square sheets of aluminium of side 42.0 cm. Determine whether or not it is realistic for Michael to produce a capacitance of $2.7 \mu\text{F}$ using these two sheets of aluminium and no dielectric. [4]

- (c) State what happens to the capacitance of a parallel plate capacitor when a dielectric is placed between the plates. [1]

Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
1	(a)	(i) Correct answer only = 14 pF (1) (ii) Correct equation used i.e. $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}$ (1) Correct answer = 1.14 pF (1) (iii) Correct use of $\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2}$ for 2 pF and 4 pF (1) Adding 8 (pF) (1) Final answer = 9.33 pF (1) (allow 1 mark for 3.43 pF or $\frac{24}{7}$) Deduct 1 mark max if answers expressed as fractions	1	1		6	5	
	(b)	Area of 0.42×0.42 seen [0.1764] (1) Rearrangement i.e. $d = \frac{\epsilon_0 A}{C}$ (1) Separation = $5.78 \times 10^{-7} \text{ [m]}$ OR $C = 15.6 \text{ [nF]}$ ish (coming from using $d = 1 \text{ mm}$) (1) Unrealistic - too thin OR capacitance too small - not achievable (1) ecf			4	4	3	
	(c)	Increases	1			1		
		Question 1 total	4	3	4	11	8	0

Electrostatic and Gravitational Fields of Force

2017

(c) Show that the electrical potential at point C is zero.

[2]

(d) A positron is initially at rest at point C and accelerates rapidly due to the electric field. Calculate the potential at the point where the velocity of the positron is $5.4 \times 10^7 \text{ ms}^{-1}$.

[3]

Question	Marking details	Marks available				Maths	Prac
		AO1	AO2	AO3	Total		
2 (a)	Arrow at C going left (1) Arrow at C going up (1)	2			2		
(b)	Use of the equation $\frac{kq}{r^2}$ (1) 52 500 000 or 39 400 000 correct (1) Applying Pythagoras $(52.5M^2 + 39.4M^2)^{0.5}$ (1) Method for obtaining angle correct e.g. $\tan^{-1}\left(\frac{525}{394}\right)$ (1) Correct answers 65.6 MV m ⁻¹ unit mark and 53 or 37[°] or 0.93 or 0.64 [rad] (1)	1	1 1 1 1		5	5	
(c)	Use of the equation $\frac{kq}{r}$ (1) $\frac{28}{8} - \frac{21}{6}$ or equivalent seen to be zero (1)	1	1		2	2	
(d)	Conservation of energy applied e.g. $eV = \frac{1}{2}mv^2$ or E_k calculated to be 1.34×10^{-15} [J] (1) 8 300 [V] (1) Negative sign (1) i.e. 8 300 [V] gets 2 marks, (-8 300 [V] 3 marks)		3		3	2	
	Question 2 total	4	8	0	12	9	0

2018

2. (a) The escape velocity, v , of a mass, m , from a spherical mass, M , and radius, R , can be calculated using:

$$\frac{1}{2}mv^2 - \frac{GMm}{R} = 0$$

- (i) Explain how this equation is an application of conservation of energy. [3]

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- (ii) Calculate the escape velocity from the Sun ($M_{\text{Sun}} = 1.99 \times 10^{30} \text{ kg}$, $R_{\text{Sun}} = 6.96 \times 10^8 \text{ m}$). [3]

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- (b) (i) The temperature of the surface of the Sun is 5780 K. Use a kinetic theory equation to show that the rms speed of a free electron on the surface of the Sun is approximately 500 km s^{-1} . [4]

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- (ii) By considering your answers to (a)(ii) and (b)(i), explain why the Sun has a slight positive charge. [2]

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- (iii) A student claims that a positive charge of approximately 0.08 C on the Sun is enough to produce an electrostatic force equal to the gravitational force on an escaping electron. Determine whether or not she is correct. [3]

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- (iv) Estimate the percentage of lost electrons compared with the total number of electrons on the Sun. Assume that the Sun is mainly hydrogen and that it has lost 0.08 C of charge in the form of electrons. [3]

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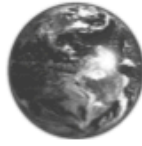
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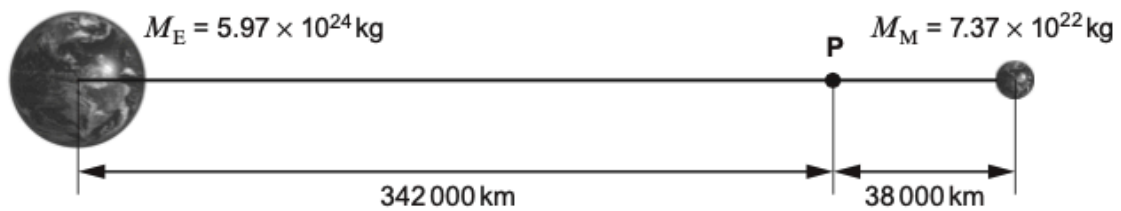
Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	(a)	(i)	Kinetic energy and [gravitational] potential energy referred to (1) $\frac{1}{2}mv^2 - \frac{GMm}{r} = \text{total [initial] energy or } \Delta KE = [-]\Delta PE$ (1) Final and initial energy are zero and equal [accept final KE = 0 or final PE = 0] (1)	3			3		
		(ii)	Rearrangement / simplification i.e. $v^2 = \frac{2GM}{r}$ (1) [accept $m = 1$ inserted] Substitution e.g. $v^2 = \frac{2 \times 6.67 \times 10^{-11} \times 1.99 \times 10^{30}}{6.96 \times 10^8}$ (1) Answer = 618 [or 620] km s ⁻¹ (1) c.a.o. [accept ~600 km s ⁻¹ b.o.d.]	1	1 1		3	3	
	(b)	(i)	Applying KE of molecule/atom/particle = $\frac{3}{2}kT$ (or deriving) (1) Rearrangement e.g. accept $v^2 = \frac{3kT}{m}$ (1) Substitution e.g. $v^2 = \frac{3 \times 1.38 \times 10^{-23} \times 5780}{9.11 \times 10^{-31}}$ (1) Answer = 513 km/s (1) [accept just ~500 km s ⁻¹ only with correct substitution]	1	1 1 1		4	3	
		(ii)	Many electrons have enough KE to escape (ecf) or just 'some electrons escape' (1) Because (b)(i) close to (a)(ii) / some electrons have higher velocity / [Boltzmann] distribution / collisions] (ecf) (1) Or protons don't escape		2		2		
		(iii)	Valid method employed e.g. electrostatic force & gravitational forces calculated/equated or other (1) Valid calculation carried out correctly e.g. $F_E = 2.4 \times 10^{-28}$ N and $F_g = 2.5 \times 10^{-28}$ N or charge = 0.084 C (1) [Or using $GMm = kQq \rightarrow 1.2 \times 10^{-10}$ [N m ²] and 1.15×10^{-10} [N m ²]] Valid conclusion (not independent) e.g. she's quite close (1) ecf from calculation of forces [not fields] or alternative above. [Accept: E force bigger / G force smaller]			3	3	2	
		(iv)	Sun electrons $\approx \frac{1.99 \times 10^{30}}{1.66 \times 10^{-27}} \approx 1.2 \times 10^{57}$ (assumption dependent, allow Sun composed of deuterium) (1) [Accept $M_S / 1$ u] $\frac{0.08}{e} = 5 \times 10^{17}$ electrons lost or charge on Sun's electrons = 1.92×10^{38} [C] (1) % lost $\approx 10^{-38}$ (1)		3		3	3	
			Question 2 total	5	10	3	18	11	0

2019

2. (a) Assuming that the Earth is an isolated perfect sphere, draw its gravitational field lines and equipotential surfaces. [3]



- (b) (i) Use the information in the diagram to calculate the gravitational potential at point P. [3]



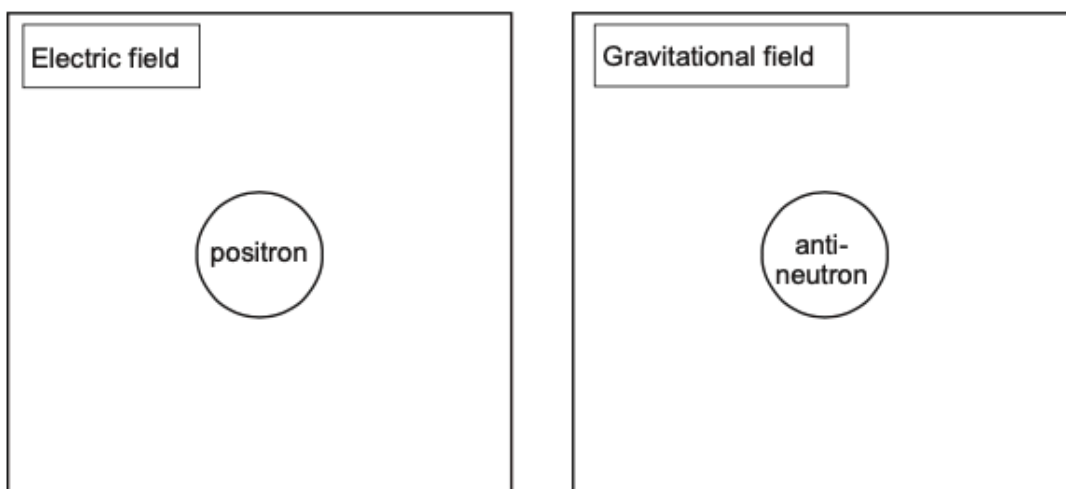
- (ii) Use the information in the diagram to show that the resultant gravitational field at point P is very small. [2]

- (iii) Myfanwy correctly calculates that the force on a 25 tonne spaceship would be negligible at point **P** and that the force would increase by approximately 0.5 N for every 10 km moved away from point **P** towards the Earth. Dafydd then concludes that the spaceship will perform simple harmonic motion about point **P**. Deduce whether or not Dafydd is correct (**no further calculations are required**). [3]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Field lines radial minimum of 3 (1) Field lines / arrows towards the Earth (1) Equipotentials are spheres minimum of 1 (circular cross-section) (1)	3			3		
	(b)	(i)	Use of $V = [-] \frac{GM}{r}$ (1) Answer = 1.16×10^6 or 1.29×10^5 (1) Answer = $-1.29 \times 10^6 \text{ J kg}^{-1}$ *** unit mark (1)	1	1 1		3	1	
		(ii)	Use of $g = [-] \frac{GM}{r^2}$ (implied by 0.0034) (1) Evidence of cancelling fields e.g. $\frac{597}{342^2} - \frac{7.37}{38^2}$ or $0.0034 - 0.0034$ or difference in fields = 1.71×10^{-7} or checking ratio of fields is equal to 1 (1)	1	1		2	1	
		(iii)	Definition of SHM or force / acceleration proportional [& opposite] to displacement (can be implied through logical argument) (1) Force is in same direction as displacement / space craft will accelerate towards Moon or Earth (1) Hence, Dafydd is wrong consistent with argument and ecf (1)			3	3		
			Question 2 total	5	3	3	11	2	0

2022

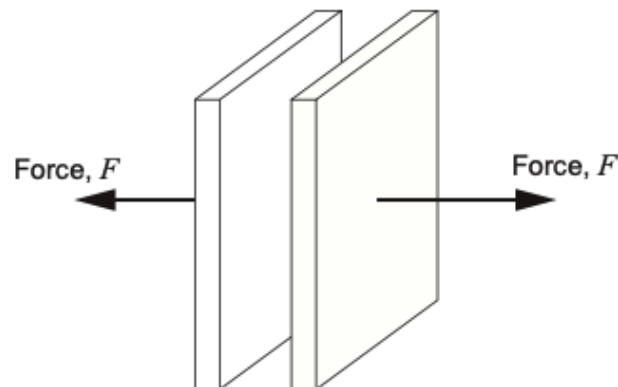
- [3]



- [6 QER]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		At least 4 field lines are radial and symmetrical (1) At least 2 equipotentials are circles / spheres (1) Out of positron and into anti-neutron (1)	3			3		
	(b)		Indicative content: Differences 1. electric fields - charges & grav fields – masses (or force per unit mass, force per unit charge or forces between charges & masses) 2. e-fields – into & out, g-fields only into (direction point – also accept grav forces only attractive, e-forces attractive & repulsive) 3. Gravity is far weaker (on small scale) OR electrical negligible on large scale (i.e. scale point) 4. E potential work done per unit charge & G work done per unit mass 5. E-fields depend on permittivity OR g-fields depend on dark matter Similarities 1. Same shape field (for point/spherical masses/charges). Note: this point is different from the same shape graph which is the next point. 2. inverse square law (applies to both) 3. Both fields lead to (concept of) PE 4. Potential or PE is inversely proportional to distance 5. Both fundamental force laws 6. Both have scalar fields OR both have vector fields (or both!) 7. Anything to do with exchange particles (photon, gluon, bosons) NOTE: some differences can be viewed as similarities and vice versa e.g. energy per unit charge and energy per unit mass could be considered a similarity. Also, the different exchange particles could be considered a difference.	6			6		

- (c) The separation of the plates of a charged capacitor is increased by the application of a force, F . The capacitor is **isolated** so the charges on the plates remain unchanged.



- (i) State why a force must be exerted to separate the charged plates. [1]

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- (ii) Explain why the capacitor stores more **energy** when the separation of the plates is increased even though the charge remains constant. [2]

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(iii) Show that the energy stored by the capacitor is given by:

$$\text{energy stored} = \frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$$

where Q is the charge stored, d is the separation of the plates, ϵ_0 is the permittivity of free space and A is the area of the plates. [2]

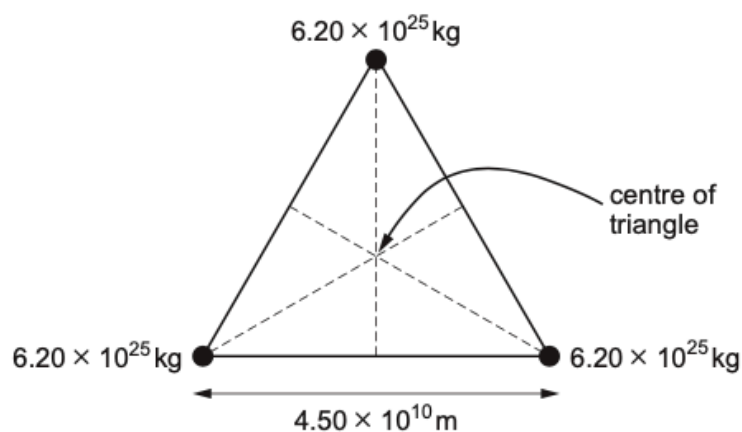
(iv) Bethan states that the force, F , required to separate the plates is given by:

$$F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$$

Determine whether Bethan is correct to arrive at this conclusion. [2]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)	Either opposite charges or attractive force [on plates]		1		1		
		(ii)	Either work done (1) and conservation of energy applied (or implied) (1) OR energy = $\frac{1}{2} \frac{Q^2}{C}$ (1) and C decreases (1) OR $\frac{Q}{V} = \frac{\epsilon_0 A}{d}$ so $V \propto d$ (1) $E = \frac{1}{2} QV$ hence correct (1) OR energy stored in field (1) field has greater volume (1) OR increased the potential energy (1) Due to attractive force (1)		2		2		
		(iii)	2 or 3 valid equations used e.g. $\frac{1}{2} \frac{Q^2}{C}$ and $C = \frac{\epsilon_0 A}{d}$ (1) Convincing working leading to $\frac{1}{2} \frac{Q^2 d}{\epsilon_0 A}$ (if they quote the 2 equations above then simply writing the equation is acceptable) (1)		2		2	2	
		(iv)	Work = force $\times$ distance quoted (1) Hence $F = \frac{1}{2} \frac{Q^2}{\epsilon_0 A}$ or distance = d and Bethan correct OR Bethan wrong at large separations or when $d = 0$ OR good conclusion well argued (1)			2	2	1	
			Question 4 total	1	8	4	13	8	0

3. Three planets of equal mass are arranged in an **equilateral triangle** as shown.



- (a) (i) Show that the gravitational force exerted by one planet on another is approximately $1.3 \times 10^{20} \text{ N}$. [2]

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- (ii) Hence, explain why the resultant gravitational force experienced by any of the planets is approximately $2 \times 10^{20} \text{ N}$ towards the centre of the triangle. [4]

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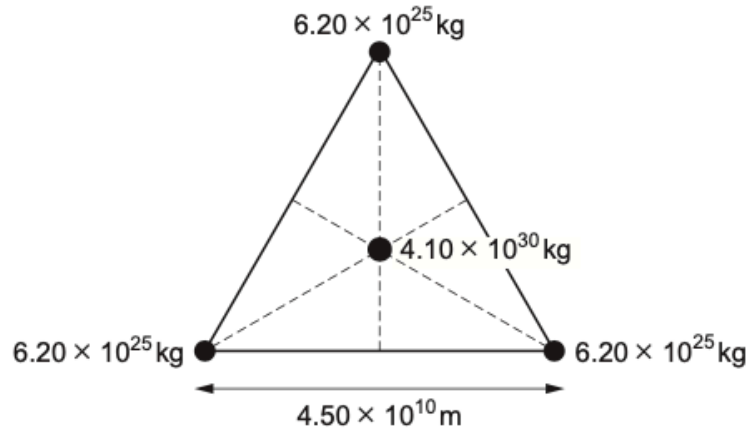
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- (b) A star is located at the centre of the triangle.



- (i) By adding to the diagram, show that the distance between the star and any of the planets is given by: [2]

$$\frac{4.5 \times 10^{10}}{2 \times \cos 30^\circ}$$

- (ii) Rhodri states that the force calculated in part (a)(ii) is negligible compared with the gravitational force exerted on a planet by the star. Determine whether or not Rhodri is correct. [2]

- (c) (i) Show that the gravitational potential at the position of a planet due to the star and the other two planets is approximately $-1 \times 10^{10} \text{ J kg}^{-1}$. [3]

- (ii) Rhodri now states that, if an object with kinetic energy $7 \times 10^{35} \text{ J}$ struck one of the planets, the planet would gain enough energy to escape the gravitational pull of the star. Discuss to what extent Rhodri's statement might be correct. [3]

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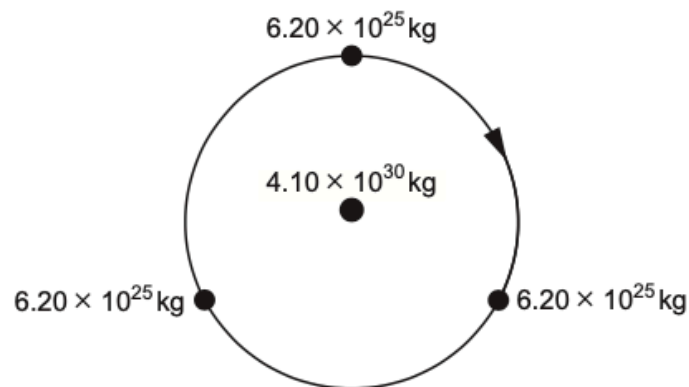
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- (d) The three planets, whilst orbiting the star, stay in an equilateral triangle. Delyth claims that these planets cannot be detected by the periodic Doppler shift of the star. Determine whether or not Delyth is correct. [3]



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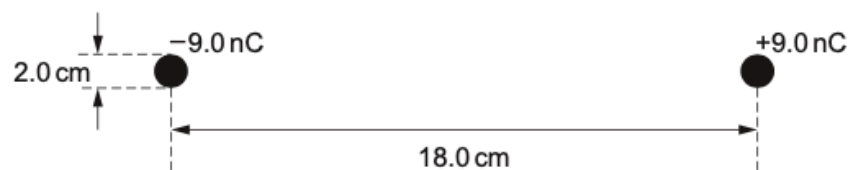
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Use of Newton's gravitational force equation (1) Answer = 1.266×10^{20} [N] or final substitution seen i.e. $\frac{6.67 \times 10^{-11} \times (6.2 \times 10^{25})^2}{(4.5 \times 10^{10})^2}$ (1)	1	1		2	2	
		(ii)	Each planet experiences two [equal and attractive] forces [from each of the other two] (1) N.B. same forces can be implied Resultant must be along centre lines (from symmetry) (1) $\times \cos 30^\circ$ seen (1) $2 \times 1.3 \times 10^{20} \times \cos 30^\circ$ (seen) OR 2.25 OR 2.19 (seen) (1)		4		4	2	
	(b)	(i)	30° angle seen or explained (1) $\cos 30^\circ = \frac{2.25 \times 10^{10}}{d}$ or equivalent seen (1)		2		2	2	
		(ii)	Mass is [far] greater (1) can be implied if substituted Factor of 10^5 or calculation 2.5×10^{25} N so Rhodri is correct (1) No penalty for incorrect distance as long as force is much bigger			2	2	1	
	(c)	(i)	Potential equation used i.e. $[-] \frac{GM}{r}$ (1) Potential due to star = $[-] 1.05 \times 10^{10}$ (1) Other potentials negligible (either stated or full calculation) (1)	1	1 1		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	PE of system / planet = $[-] 1 \times 10^{10} \times 6.2 \times 10^{25} = (6.2 \times 10^{35} \text{ J})$ (1) Alternative for the 1st mark: Obtaining escape velocity: $140\,000 \text{ [m s}^{-1}\text{]}$ and velocity of $7 \times 10^{35} \text{ [J]}$ i.e. $150\,000 \text{ [m s}^{-1}\text{]}$ Another alternative for the 1st mark: $\frac{7 \times 10^{35}}{6.2 \times 10^{25}} = 1.13 \times 10^{10} \text{ [J kg}^{-1}\text{]}$ However, conservation of momentum / other energy transfers must be considered (1) minimum required is {elastic / inelastic} collision Conclusion consistent with their argument but must be based on a value of PE or potential or escape velocity (1)						
	(d)		Star is at centre of mass or resultant force will be zero (1) Hence, no wobble / orbit etc. or stationary (1) No Doppler shift and not detectable so Delyth is correct (1)			3	3	1	
			Question 3 total	2	9	8	19	10	0

6. Two small spheres of radius 1.0 cm carry uniform charges of +9.0 nC and -9.0 nC. Their centres are separated by a distance of 18.0 cm as shown. Treat the charges as being at the centres of the spheres.



- (a) (i) Show that the potential at the surface of the negative sphere is approximately -7500 V. [3]

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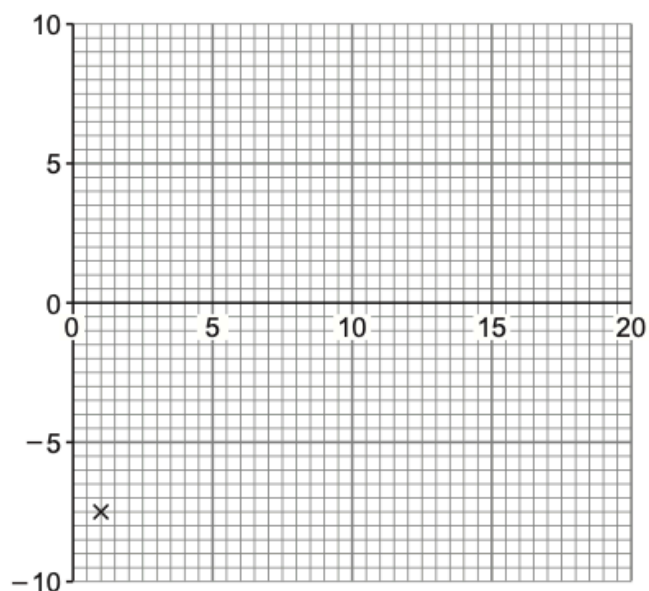
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- (ii) Sketch the graph for the electric potential between the two charges (the potential has been plotted for the surface of the negative charge). [3]

Potential / kV



Distance from
centre of
negative
charge / cm

- (b) Explain why the electric field on the right-hand surface of the negative sphere is approximately $810\,000\text{ N C}^{-1}$ and directed towards the left. [3]

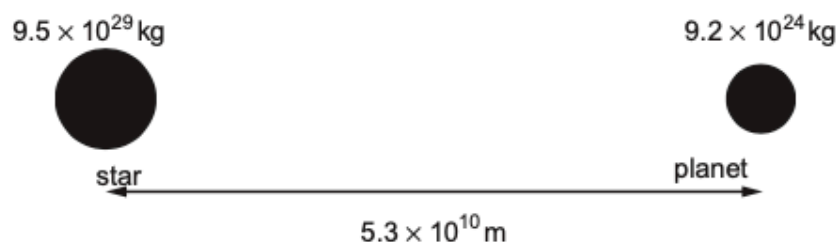
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	Use of potential equation i.e. $V = k \frac{q}{r}$ (1) -8100 [V] OR 426-506 [V] seen or implied (1) Both added / subtracted to give correct answer i.e. 7600 [V] (1)	1	1 1		3	2	
		(ii)	General shape (1) +7500 @ 16 - 18 cm (1) 0 @ 8 - 10 cm (1) N.B. Ignore fields inside charges should be 0		3		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		Use of field equation i.e. $E = k \frac{q}{r^2}$ (1) 810 000 and negligible OR 2 800 OR 2 500 seen (1) Both fields to left at this point OR lines into -ve and out of +ve (1)	1	1 1		3	2	
			Question 6 total	2	7	0	9	6	0

Orbits and the Wider Universe

2017

3. The planet (Kepler-186f) orbits the star (Kepler-186) and is considered to be the most Earth-like planet yet discovered. Approximate details of the system are shown in the diagram below.



- (a) (i) Show that the radius of orbit of the star around the centre of mass of the system is 513 km. [2]

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- (ii) The period of orbit of the planet is 130 days. Calculate the orbital velocity of the star. [3]

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- (iii) The velocity calculated in part (a)(ii) is small. Explain why small velocities are difficult to measure using red shift. [2]

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- (b) (i) Calculate the gravitational force between the star and the planet. [2]

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- (ii) An historic alternative to Newton's Law of Gravitation involved invisible steel rods connecting stars and planets and providing the force for keeping planets in orbit. A hypothetical cylindrical steel rod of radius $7.0 \times 10^6 \text{ m}$ and length $5.3 \times 10^{10} \text{ m}$ joins the planet to the star. Determine whether or not this steel rod is strong enough to keep the planet in orbit and explain why this theory is not accepted. (Breaking stress of steel = $4.5 \times 10^8 \text{ Pa}$.) [4]

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- (c) (i) Calculate the potential energy of the star-planet system (see diagram in part (a)). [2]

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- (ii) In fact the planet has a slightly elliptical orbit. Explain how conservation of energy applies to this elliptical orbit. [2]

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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)	(i)	Use of the equation $\frac{M_2 d}{(M_1 + M_2)} = x$ even if wrong M_2 (1) $\frac{9.2 \times 10^{24} \times 5.3 \times 10^{10}}{(9.5 \times 10^{24} + 9.2 \times 10^{24})}$ seen (1) ($\{ +9.2 \times 10^{24} \}$ can be omitted from the above equation) or 513 263 [m] seen etc.	1 1			2	2	
		(ii)	$v = \frac{2\pi r}{T}$ applied or $v = \omega r$ and $\omega = \frac{2\pi}{T}$ combined (1) Conversion of 130 days ($130 \times 24 \times 3600$) (1) Answer = $0.287 [\text{m s}^{-1}]$ (ecf on day conversion) (1)	1	1 1		3	3	
		(iii)	Redshift is small or the redshift is proportional to velocity or small $\Delta\lambda$ or shown using the Doppler equation (1) Small wavelength change is difficult to measure or shifted wavelength is too close to the original (1)		2		2		
	(b)	(i)	$F = \frac{GMm}{r^2}$ used (1) Answer = $2.08 \times 10^{23} [\text{N}]$ (1)	1	1		2	2	
		(ii)	Method for obtaining stress i.e. $\frac{F}{A}$ (1) $1.35 \times 10^9 [\text{Pa}]$ (1) Correct conclusion e.g. the steel bar would break ecf (1) Theory doesn't work / Newton's grav law used successfully for centuries / won't work for elliptical orbits / would melt (1) Alternative: Use of $F = \sigma A$ (1) $F = 6.9 \times 10^{22} [\text{N}]$ (1)			4	4	2	

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(c)	(i)	$\text{PE} = -\frac{GMm}{r}$ used N.B. ecf if $\text{PE} = Fr$ used (1) Answer = $[-]1.10 \times 10^{34} [\text{J}]$ (1)	1	1		2	2	
		(ii)	Realising KE is involved (1) Correct variation with distance for KE or velocity or as PE increases KE decreases or converse (1)			2	2		
			Question 3 total	5	6	6	17	11	0

2018

3. (a) Explain why the age of the Universe can be approximated as $\frac{1}{H_0}$, where H_0 is the Hubble constant. [2]

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- (b) The megaparsec (Mpc) is a unit of astronomical distance equal to 3.09×10^{22} m. Use Hubble's law to show that the expected redshift for a supernova at a distance of 1 Mpc for a wavelength of 486.1 nm is approximately 0.11 nm. [3]

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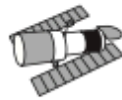
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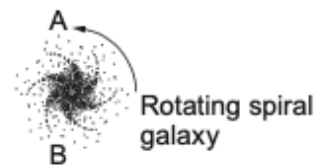
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- (c) The spiral galaxy shown is rotating anticlockwise and is viewed by the Hubble Space Telescope.

Hubble Space
Telescope



Images not to scale



The measured **blue** shift at point A of the galaxy is 0.22 nm (for the 486.1 nm wavelength) and the measured **redshift** at point B of the galaxy is 0.66 nm (for the same wavelength). Calculate the recessional velocity of the galaxy and the rotational speed of the galaxy at A (assume that A and B have the same rotational speeds). [5]

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- (d) Explain how spiral galaxies provide evidence for the existence of dark matter and how this evidence has been gathered. [6 QER]

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
3	(a)		Constant velocity (1) Substitution of $D = vT$ into Hubble's law and convincing algebra (1)	2			2	1	
	(b)		Substitution of H_0 & 3.09×10^{22} m into Hubble's law (gives 68 000 m/s) (1) Substitution [of at least c and λ] into Doppler equation (1) $\Delta\lambda = \frac{68\,000 \times 486.1 \times 10^{-9}}{3 \times 10^8}$ seen (1) or 0.11 nm	1 1	1		3	3	
	(c)		Correct method for obtaining velocities e.g. $\frac{c\Delta\lambda}{\lambda}$ (1) Blue shift speed is $-12v$ (-136 000) (1) and redshift speed is $6v$ (407 000 / 408 000) (1) Recessional speed = $2v$ (136 000) (1) Rotational speed = $4v$ (272 000) (1) NB Final mark awarded only if recessional speed and rotational speed clearly identified.		5		5	5	
	(d)		Collation points C1 Doppler/red/blue shift measured C2 of [known] lines / wavelengths C3 velocity calculated / measured / linked to $\Delta\lambda / \frac{\Delta\lambda}{\lambda} = \frac{v}{c}$ C4 for various distances from centre of galaxy [can be credited from good graph] C5 absorption spectrum used C6 mention of [large] telescope / analysis of light / em radiation observed C7 mention of spectrometer / prism / grating / spectral analysis Theory points T1 Doppler shift explained e.g. redshift, moving away T2 Doppler equation quoted or shift dependency on velocity mentioned T3 Orbital speed increases with mass / $v = \sqrt{\frac{GM}{r}}$ T4 Orbital speed decreases with radius (theoretically) OR graph	6			6		

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
			or equation T5 Theoretical speed based on visible/baryonic mass OR graph Results & Conclusion points R1 Speed does not decrease with radius / actual speed too large OR graph R2 Hence extra mass R3 Dark matter linked to extra mass R4 Possible link to Higgs boson / WIMPS / CMBR						

2019

3. (a) (i) State Kepler's 1st and 2nd laws.

[3]

(ii) Kepler's 3rd law can be derived from Newton's gravitational law and the equation for centripetal motion. Show that, for any object in a circular orbit about the Earth:

$$T^2 = \frac{4\pi^2}{GM_E} r^3$$

where T = the period of orbit, r = the radius of orbit, G = the gravitational constant and M_E = the mass of the Earth. [3]

- (b) The radius of the geostationary orbit above the Earth's equator is 42000km and the Earth-Moon distance is 380000km. Use Kepler's 3rd law to calculate the period of the Moon's orbit. [3]

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- (c) Estimate the minimum possible orbital period for a satellite orbiting the Earth, stating any assumption that you make (the radius of the Earth is 6370 km). [3]

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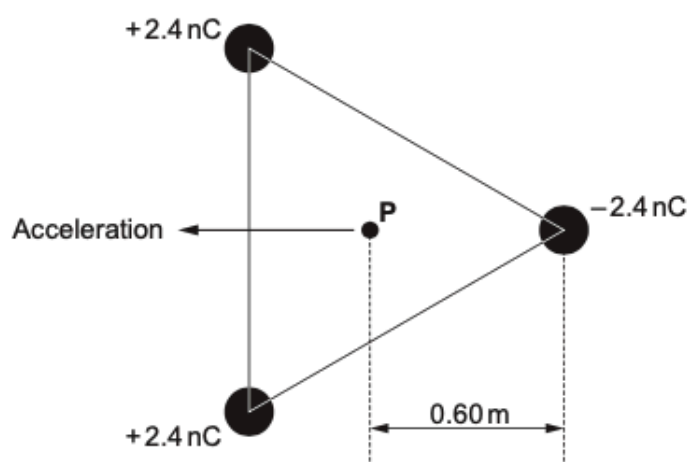
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
3	(a)	(i)	Planets / masses orbit in ellipses (with the Sun at a focus) (1) [Line from planet to Sun] sweeps equal area(s) in equal time(s) or sweeps area at a constant rate (1) Sun at a focus & line from planet to Sun (or radius vector) (1)	3			3		
		(ii)	$\frac{GMm}{r^2} = \frac{mv^2}{r}$ or $mr\omega^2$ (1) Substitution of $\omega = \frac{2\pi}{T}$ or equivalent i.e. $v = \frac{2\pi r}{T}$ (1) Intermediate step seen e.g. $\frac{GMm}{r^2} = m \frac{4\pi^2}{T^2} r$ (1) Alternative: $T = 2\pi \sqrt{\frac{d^3}{G(M_1 + M_2)}}$ (1) $M_2 = 0$ or implied (1) $d = r$ and $M_1 = M_E$ (1)	3			3	3	
	(b)		Period of geostationary = 1 day (24 hr etc.) (1) Use of Kepler e.g. $\frac{T_1^2}{r_1^3} = \frac{T_2^2}{r_2^3}$ [accept substitution with 5.97×10^{24}] (1) Correct answer = 27.2 [days] (2.35×10^6 s) (1) Alternative: Substitution into period equation (1) Algebra (1) Correct answer = 27.2 [days] (2.32×10^6 s) (1)	1 1	1		3	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Assumption - Radius cannot be less than r_E (or inside Earth's surface etc.) or ignore mountains/air resistance accept no atmosphere or Earth is totally smooth (1) Use of Kepler or accept equation use again (1) Correct answer = 1.42 h, $\frac{1}{16.9}$ of a day, 5 100 s etc. (accept slightly larger due to, e.g. $r_E = 6470$ km to be above atmosphere etc. which gives 1.45 h) (1)		3		3	2	
			Question 3 total	8	4	0	12	7	0

2022

3. Three stationary charges are located at the corners of an **equilateral** triangle and an electron is located at the centre, **P**, of the triangle as shown. The electron is exactly 0.60 m from each charge.



- (a) Draw arrows to show the 3 electric fields at **P** due to the 3 charges. Use these to explain why the electron will accelerate in the direction shown. [4]

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(b) Calculate the initial acceleration of the electron.

[4]

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(c) Show **clearly** that the initial potential energy of the electron is approximately $-6 \times 10^{-18} \text{ J}$.

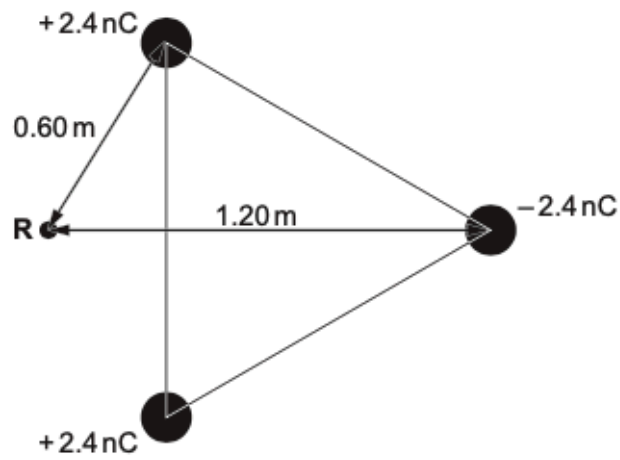
[2]

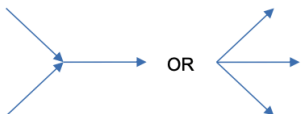
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- (d) Determine whether or not the electron (initially at rest) will reach point R, shown in the diagram below (both positive charges are 0.60 m from R). Explain your reasoning. [4]



Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
3	(a)	Vector diagram roughly as shown (1)  Vertical components cancel (1) Accept forces due to +ve charges cancel Field is right OR attractive force towards positives (1) Electron is negative linked to F (or a) left OR repulsive from -ve (1)		4		4	2	
	(b)	Use of $F = \frac{kQq}{r^2}$ OR $F = EQ$ and $E = \frac{kq}{r^2}$ (1) (expect to see 9.6×10^{-18} [N]) $2E\cos 60 + E$ OR $2E\sin 30 + E$ OR $2F\cos 60 + F$ OR $2F\sin 30 + F$ (1) (expect to see 1.92×10^{-17} [N] or 120 [N C⁻¹]) Use of $F = ma$ (1) (can be awarded for any valid force $\div 9.11 \times 10^{-31}$) Correct answer = 2.1×10^{13} [m s⁻²] (no ecf for this, must be correct) (1)	1					
			1	1		4	3	
				1				

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Substitution into $PE = \frac{kQq}{r}$ e.g. $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times 1.6 \times 10^{-19}}{0.6}$ (1) $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6} + \frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6} +$ $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times 1.6 \times 10^{-19}}{0.6}$ seen or equivalent or $\frac{9 \times 10^9 \times 2.4 \times 10^{-9} \times -1.6 \times 10^{-19}}{0.6}$ seen with statement that the other 2 potentials cancel (1)	1					
					1		2	1	
	(d)		Valid method e.g. considering potential energy/potential (1) New PE = $2 \times -5.76 \times 10^{-18} + 2.88 \times 10^{-18} = -8.64 \times 10^{-18}$ [J] OR new potential = 54 [V] OR PE must be lower because -ve values are the same but the positive is smaller (doesn't need a calculation) OR new potential is higher than original because +ve potentials are the same but negative is smaller (36V / 37.5 [V] to 54 [V]) (1) Can travel to lower PE OR will gain KE OR can travel to higher potential because electron is negative (1) Correct conclusion based on correct physics (yes is sufficient if the explanation is clear) (1) Alternative based on forces: Attractive forces (or forces due to +ve charges) are the same/symmetrical for both points OR oscillating with SHM if attractive forces only OR would just reach R with attractive forces only OR force at R is 7.2×10^{-18} [N] ($\rightarrow$) (1) Repulsive force / force due to negative charge helps OR repulsive force at end is 2.4×10^{-18} [N] OR force at P is 19.2×10^{-18} [N] ($\leftarrow$) (1) Overall of these is to the left (1) Conclusion – yes (1)			4	4	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
			Award 3 marks for: Yes it gets there with 16.5 / 18 eV of KE (with no reasoning) Yes it gets there with 2.88 / 2.64×10^{-18} J of KE (with no reasoning) Yes it gets there with a speed of 2.5×10^6 m s ⁻¹ (with no reasoning) Award 2 marks for: Yes it gets there because the force due to -2.4 nC always repels Yes because acceleration left is greater than acceleration right Award 1 mark for: Yes / don't know / No because positive charges provide force to right at the end.						
			Question 3 total	3	7	4	14	8	0

5. (a) (i) Derive the expression for the critical density of a flat universe. [4]

$$\rho_c = \frac{3H_0^2}{8\pi G}$$

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(ii) Use this equation to show that the critical density of the universe corresponds to approximately 5 hydrogen atoms per m³. [2]

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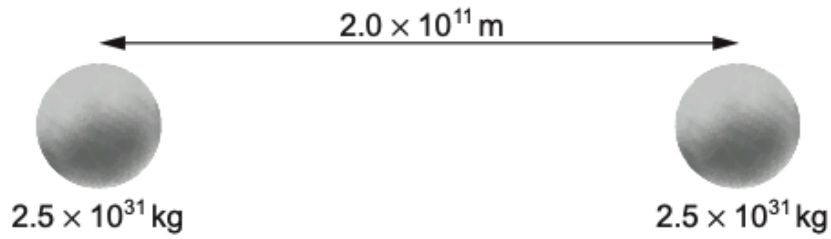
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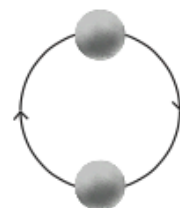
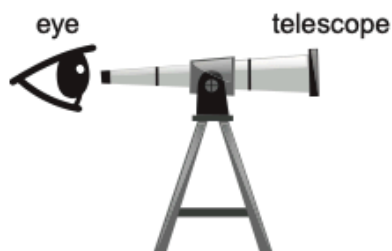
- (b) A two-star system has two stars of equal mass as shown.



- (i) State or calculate the position of the centre of mass of the two-star system. [1]

- (ii) Calculate the period of orbit of the stars about their centre of mass. [2]

- (iii) Calculate the maximum red shift (or blue shift) of the hydrogen 434 nm line due to the orbital motion of the stars. [3]



Not drawn to scale

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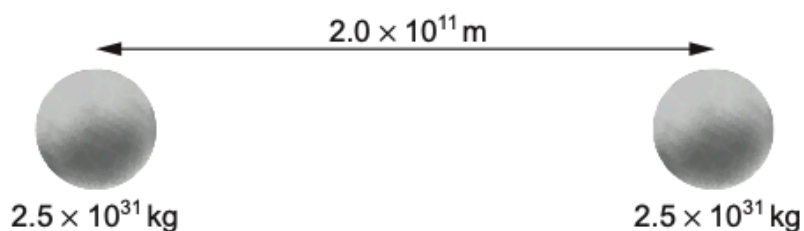
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- (c) The centre of mass of this two-star system is also the point where the resultant gravitational field strength is zero. Explain why these points only coincide when the stars have identical masses. [2]



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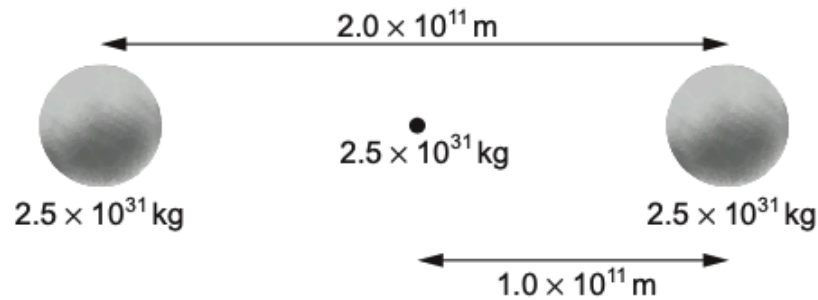
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- (d) A different star system consists of three stars all of equal mass. Two stars orbit around a stationary black hole and the black hole is always halfway between the two stars as shown.



- (i) Explain why the resultant force on the black hole is always zero. [1]

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- (ii) Explain why the gravitational force acting on either of the orbiting stars is five times greater in this three-star system than the two-star system of part (b). [2]

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- (iii) Joseff claims that the stars in the three-star system will provide a red shift that is five times larger than that of the two-star system of part (b). Evaluate whether or not he is correct. [3]

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- (e) A recent theoretical publication suggests that the decay of the Higgs Boson will give direct evidence for dark matter. Suggest what needs to be done for this theory to be generally accepted by scientists in the future. [2]

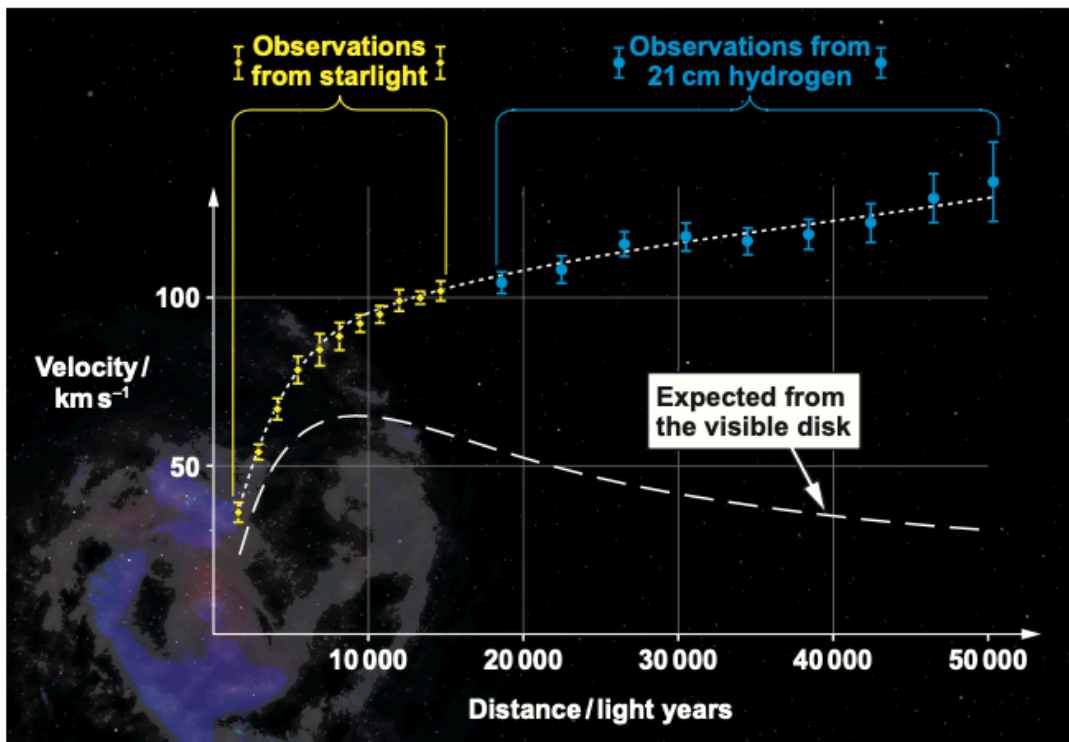
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)	(i)	KE + gravitational PE = 0 & context explained briefly (also accept KE = gravitational PE and context) (1) Minimum context = conservation of energy or mention of galaxy or spherical universe or uniform universe or cosmological principle or explaining why $r = D$ or explaining what the mass of the sphere is (accept whole universe or part and mass inside) Reference to universe nor expanding now loses 1 st mark $v = H_0 D$ substituted for v (1) mass = density $\times$ volume of sphere (1) Algebra set out clearly (1)	4			4	3	
		(ii)	Correct substitution into formula (gives 8.66×10^{-27}) (1) Dividing by 1.66×10^{-27} (or 5.2 or 5.22 seen) OR $5u = 8.3 \times 10^{-27}$ (1)	1	1		2	1	
	(b)	(i)	Halfway between or correct substitutions into C of M equation leading to 1×10^{11} or simply 1×10^{11} [m]		1		1		
		(ii)	Substitution into period equation (1) 9.73×10^6 s (allow 1 mark for 13.8×10^6 OR 4.87×10^6 OR 3.44×10^6) (1)	1	1		2	2	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Use of $v = \frac{2\pi r}{T}$ or equivalent with $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) (expect $65\,000 \text{ m s}^{-1}$) Use of Doppler equation (1) Correct answer = 0.0934 nm (accept 1sf) (1) Note: ecf on previous values gives 0.187 nm , 0.264 nm and 0.066 nm (for full marks unless another clear mistake seen) Allow 2 marks max if d used instead of r (gives 0.187 nm) Allow 1 mark max if $\frac{1}{2}mv^2 = \frac{GMm}{r}$ (also gives 0.187 nm)	1 1	 1		3	3	
	(c)		Centre of mass is nearer more massive star (1) Which will give a larger field OR but must be nearer smaller star for fields to match (1) Accept equivalent reverse argument		2		2	2	
	(d)	(i)	Force [due to both stars] are always equal [and opposite] Accept this is the point where the field is zero Accept masses of equal mass and distance either side		1		1		
		(ii)	Force due to black hole is 4 $\times$ greater or $4.17 \times 10^{30} \text{ [N]}$ (1) Also force from other star (gives factor 5) or $1.04 \times 10^{30} \text{ [N]}$ (1)		2		2	1	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Increasing v increases redshift OR linking orbital speed to redshift (can be implied by good answers) (1) Correct realisation that factor is $\sqrt{5}$ or that v^2 proportional to force (this implies the 1st mark) (1) Joseff is wrong (based on some valid physics) (1) e.g. red shift increases with orbital speed but they aren't proportional (force and speed) so Jo is wrong (this gets 2 marks – 1st and 3rd) Also accept for 2 marks: Red shift increases with orbital speed and orbital speed is proportional to force, so Jo is right Also accept calculations: Expected $v = 144\,000\text{ m s}^{-1}$ (1) Red shift = 0.21 n[m] (1) Joseff is wrong (1)			3	3	2	
		(e)	[More] experiments involving Higgs [decay] (1) Results [need to be] in agreement with theory (1)			2	2		
			Question 5 total	8	9	5	22	12	0

2023

2. Light from a spiral galaxy is analysed and the following graph of velocity against distance from the centre of the galaxy is obtained.



- (a) Show that the speed, v , of an object in a circular orbit of radius, r , about a massive object of mass, M , is given by:

[3]

$$v = \sqrt{\frac{GM}{r}}$$

- (b) Hence, explain why the graph is considered to be evidence for dark matter. [3]

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- (c) The lower curve in the graph suggests that an object a distance of 20 000 light years from the centre of the galaxy should have an orbital speed of approximately 50 km s^{-1} . Use this data to estimate the visible mass of the galaxy. (1 light year = $9.46 \times 10^{15} \text{ m}$) [3]

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- (d) The observed data at 30 000 light years from the galactic centre were obtained using microwaves of wavelength 21 cm. Calculate the approximate wavelength shift that was observed to obtain the data plotted at 30 000 light years. [3]

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- (e) Use the Hubble equation to calculate the distance, from Earth, for a galaxy to have the same recessional speed as that of part (d). [2]

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Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
2	(a)	Correct expression for centripetal force used (accept acceleration if equated to gravitational field strength) (1) accept with ω^2 Equating to gravitational expression i.e. $\frac{mv^2}{r} = \frac{GMm}{r^2}$ or $\frac{v^2}{r} = \frac{Gm}{r^2}$ (1) Enough algebra shown. Minimum is this: $\frac{mv^2}{r} = \frac{GMm}{r^2} \quad v^2 = \frac{GM}{r} \quad (1)$ Alternative 1: $m\omega^2 r$ or $\omega^2 r$ quoted (1) $m\omega^2 r = \frac{GMm}{r^2}$ and $v = \omega r$ or acceleration equivalent (1) Minimum algebra leading to answer (1) Alternative 2: Using M_1 or M_2 negligible in $T = 2\pi\sqrt{\frac{d^3}{G(M_1+M_2)}}$ (1) e.g. $2\pi\sqrt{\frac{d^3}{G(M_1+M_2)}} \approx 2\pi\sqrt{\frac{d^3}{GM}}$ Quoting $v = \omega r$ and $\omega = \frac{2\pi}{T}$ (1) or equivalent All algebra ok leading to the answer (1)	3			3	2	
	(b)	Measured velocities are too large (1) Accept velocity increasing on graph Link to the equation e.g. velocity increases with mass (1) Dark matter accounts for increased mass or the mass must be bigger (1)	3			3		

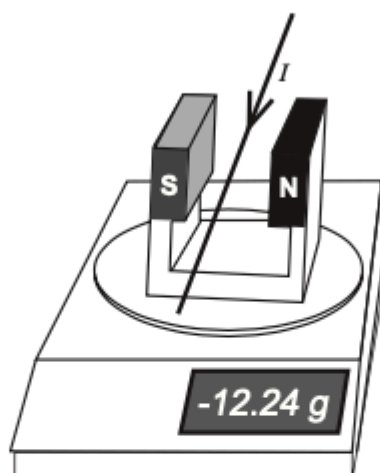
Question		Marking details	Marks available					
			AO1	AO2	AO3	Total	Maths	Prac
	(c)	Rearrangement for M i.e. $M = \frac{v^2 r}{G}$ (1) Conversion of distance i.e. $20\,000 \times 9.46 \times 10^{15} = (1.892 \times 10^{20})$ (1) Answer = 7.09×10^{39} k[g] (1)		3		3	3	
	(d)	Speed estimated as 110 - 130 [km s⁻¹] (1) Use of the Doppler equation ecf (1) Answer = 7.7×10^{-5} to 9.1×10^{-5} [m] (1)	1	1		3	2	
	(e)	Use of Hubble equation $v = H_0 d$ (1) Answer = 5.0×10^{22} to 5.9×10^{22} [m] (ecf on v) (1)	1	1		2	2	
		Question 2 total	8	6	0	14	9	0

Magnetic Fields

2017

4. A student carries out an investigation. She uses a top pan balance to investigate the relationship $F=BI\ell$ using the set up shown in the diagram. The current-carrying wire is placed perpendicular to the magnetic field, B .

The current in the wire is varied using a variable power supply while the magnetic field, B , and the length of wire in the field, ℓ , are kept constant. Before switching on the current, the balance is set to zero.



- (a) (i) Explain why the reading on the balance is negative for the direction of current and magnetic field shown in the diagram. [3]

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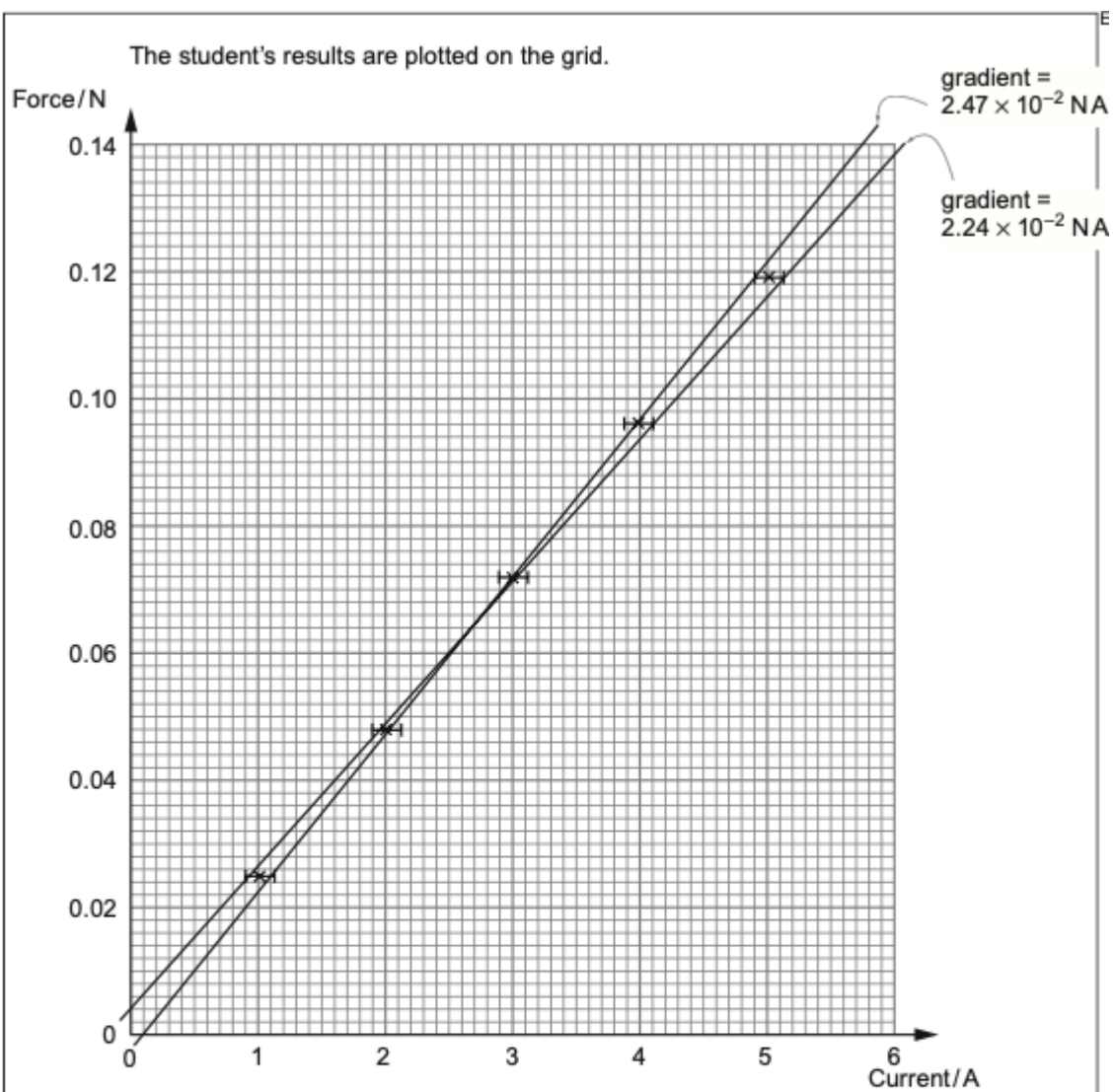
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- (ii) Calculate the mean value of the gradients ($2.47 \times 10^{-2} \text{ NA}^{-1}$ and $2.24 \times 10^{-2} \text{ NA}^{-1}$) and its **percentage** uncertainty. [3]

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- (iii) The length of wire in the uniform magnetic field is (8.0 ± 0.5) cm. Determine a value for the magnetic field strength and its **absolute** uncertainty, stating both to an appropriate number of significant figures. [4]

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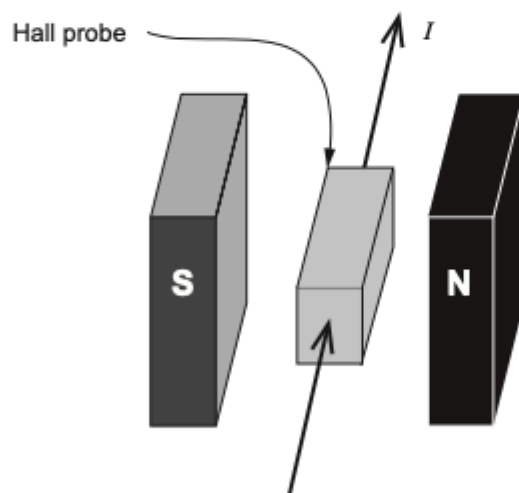
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- (b) In another experiment a Hall probe is used to confirm the value of the magnetic field strength obtained. The value of B obtained using the Hall probe is (310 ± 8) mT.



- (i) Explain how the Hall voltage arises in the Hall probe shown. [3]

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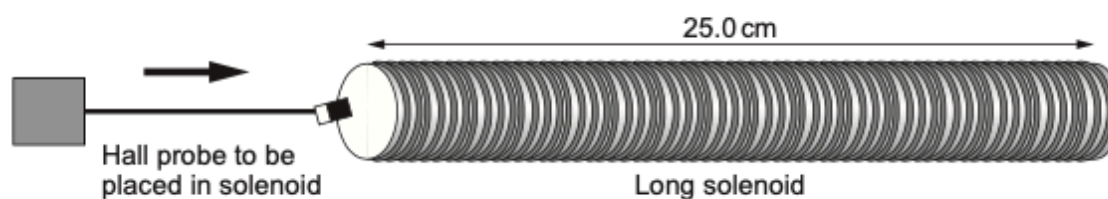
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- (ii) Discuss the effectiveness of the experiment in part (a) for confirming the relationship $F = BIl$ and measuring the magnetic field strength, compared with using the Hall probe {see (a)(iii) and (b)}. [6 QER]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Force is upward <u>on the magnet</u> (1) [F]LHR quoted and application of N3 (1) Force is down on the wire (1)	1	1 1		3		3
		(ii)	Mean calculated = 0.0236 or 0.02355 (1) accept 2.4×10^{-2} Absolute uncertainty calculated or implied (± 0.0012) (1) % uncertainty = 5% (1) accept 1 or 2 sig figs accept range 4 – 5.1%		3		3	3	3
		(iii)	Gradient = BI used or implied (1) $B = 0.295$ [T] (no sig fig penalty) (1) Adding % uncertainties ecf from (a)(ii) (11%) (no sig fig penalty) (1) accept first principles approach (0.295 ± 0.032)T or (0.30 ± 0.03) ecf must have consistency of sig figs either 1 or 2 sig figs in uncertainty + UNIT or equivalent in mT (1)		4		4	4	4
	(b)	(i)	Force on charge carriers due to mag field or [F]LHR (1) Charge carriers go to one side of conductor / reference to build-up of charge (1) Electric field set up or top is +ve / -ve depending on carriers (1)	3			3		3
		(ii)	Indicative content: 1. Line goes through origin. 2. Graph is a straight line. 3. Line passes through all the error bars. 4. Both values of B compared. 5. Hall probe is more precise or its uncertainty is smaller [accept accurate]. 6. Sensible comment comparing methods e.g. method A is graphical, method B required only one measurement or converse, method A is more prone to human error, method B doesn't confirm $F = BIl$			6	6	3	6

4. An experiment is carried out to measure the magnetic field in a long solenoid as the current in the solenoid is varied.

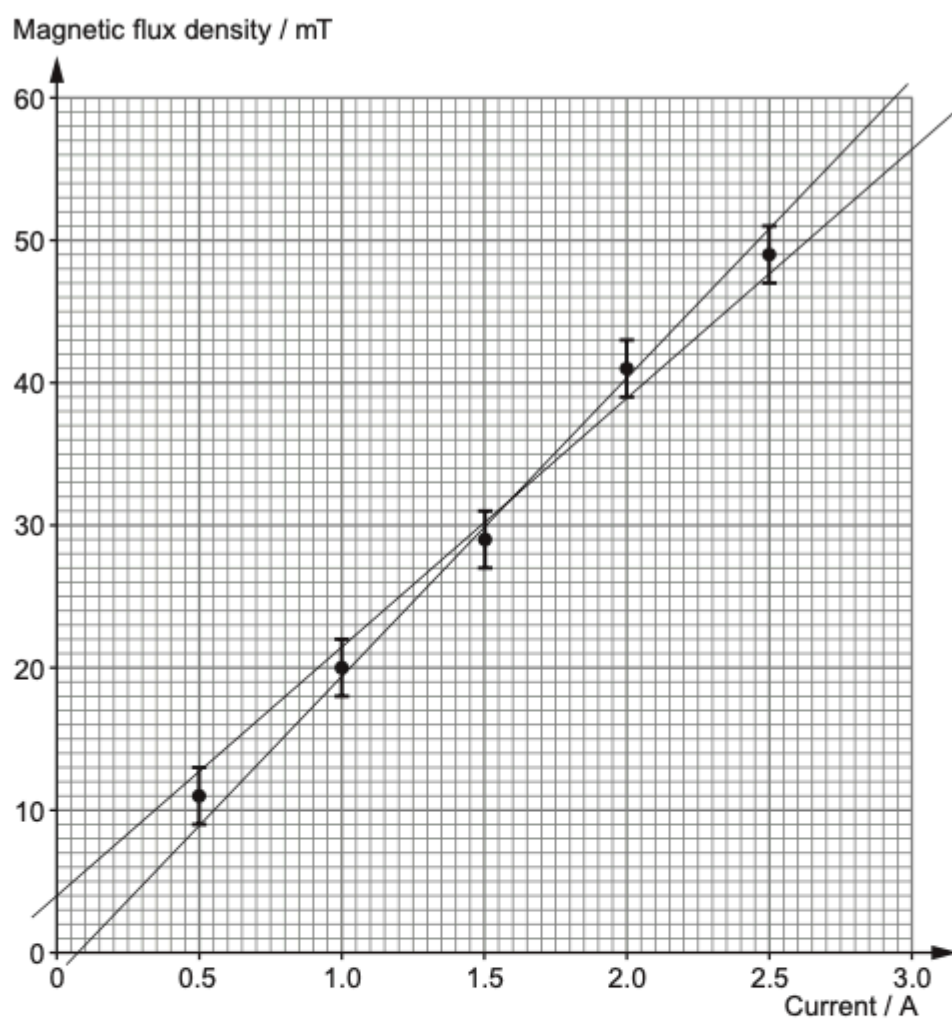


Theory states that the relationship between magnetic flux density, B , at the centre of the solenoid and current, I , is given by the equation:

$$B = \mu_0 n I$$

The results obtained are shown in the table and plotted on the grid opposite along with error bars and lines of maximum and minimum gradient.

Current / A ± 0.01 A	Magnetic flux density / mT ± 2 mT
0.50	11
1.00	20
1.50	29
2.00	41
2.50	49



(a) Calculate the mean value of the gradient along with its **percentage** uncertainty. [4]

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- (b) Calculate the number of turns in the solenoid along with its **absolute** uncertainty. The percentage uncertainty in the length of the solenoid is 0.4 %. [3]

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- (c) (i) The solenoid manufacturer states that there are exactly 5 000 turns in the solenoid. Evaluate the accuracy of your value obtained in part (b) and whether or not the graph is in agreement with the equation: $B = \mu_0 n I$. [4]

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- (ii) Suggest a reason for the disagreement between the manufacturer's stated value (5000 turns) and your value calculated in part (b). Suggest how the experimental technique might be improved for better agreement. [2]

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Valid method for obtaining gradient (implied if correct) (1) Max = ~ 0.021 or min = ~ 0.017 (1) [OK to give 21.0 & 17.5 mT A ⁻¹] Mean correct (~ 0.019) (1) Uncertainty = 8.5 % – 11% (1) [no s.f. penalty]		4		4	4	4
	(b)		Gradient = $\mu_0 n$ used or implied ($n = 15200$) (1) $N = 3780$ (1) $N = 3800 \pm 400$ (1) [allow 1 or 2 s.f. in uncertainty 1 more s.f. in N]			3	3	3	3
	(c)	(i)	5000 is outside uncertainty range [or equiv.] (1) Straight line (1) Through all error bars (1) Lines straddle the origin (1) [allowthrough origin]			4	4		4
		(ii)	Flux density least accurate/probe not in centre [end effect] /not aligned [90° to axis](1) Get better Hall probe with higher [smaller / better] resolution / put in centre/align properly (1) [not: repeat readings / better apparatus]			2	2		2
			Question 4 total	0	4	9	13	7	13

2019

4. (a) The current in a long wire is 290 A and is too great to be measured by an ammeter. Explain how you could use a Hall probe, calibrated in tesla (T), to determine this current. [3]



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- (b) Another long wire carrying the same large current is placed parallel to the original wire as shown. Calculate the force per unit length on each wire also stating the direction of the force on each wire. [4]



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- (c) (i) Tirion claims that an electron halfway between the wires, travelling at a speed of $2.9 \times 10^7 \text{ m s}^{-1}$ parallel to the wires will perform perfect circular motion between the wires. Determine, using a suitable calculation, whether or not Tirion's claim is correct. The magnetic flux density halfway between the wires is 4.64 mT . [4]



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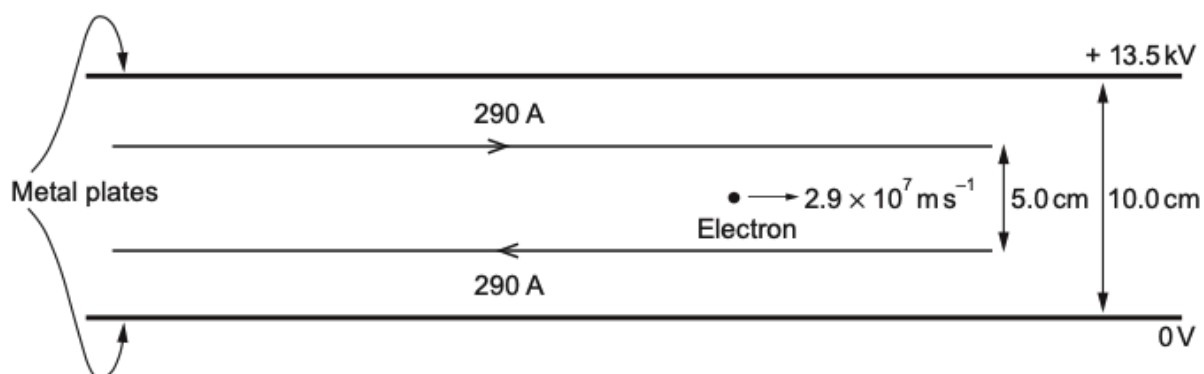
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- (ii) Sketch the motion of the electron. [2]



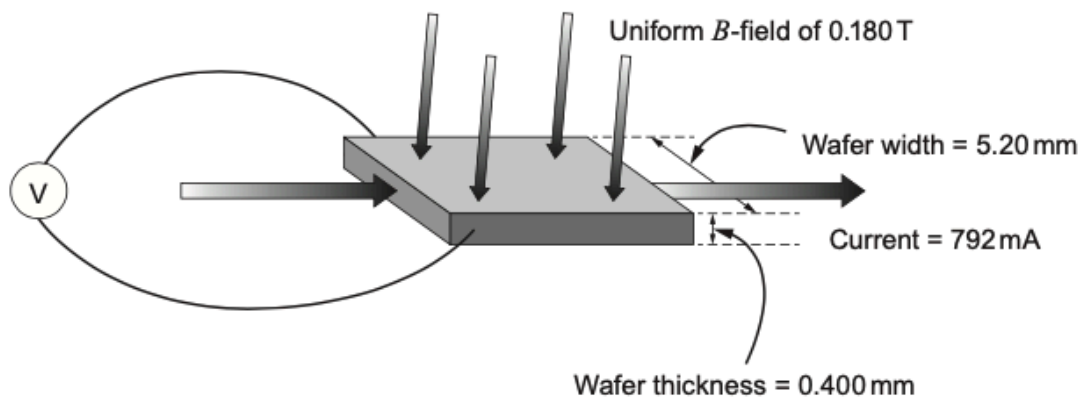
- (d) Suppose that an electron travels in a region with a magnetic field and an electric field due to two parallel metal plates as shown below. Deduce whether or not the electron continues with constant velocity ($B = 4.64 \text{ mT}$). [5]



Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Align probe for max signal / perpendicular to field (at known distance) (1) Put probe at a fixed/known distance (1) Current calculated from equation $B = \frac{\mu_0 I}{2\pi a}$ (1)	3			3	1	1
	(b)		Use of $F = BIl$ (1) Use of $B = \frac{\mu_0 I}{2\pi a}$ (1) $\left[\text{or } \frac{F}{l} = \frac{\mu_0 I^2}{2\pi a} \text{ for 2 marks} \right]$ Answer = $0.336[4 \text{ N m}^{-1}]$ (1) Repulsive force (or equivalent) (1)	1 1	1 1		4	3	
	(c)	(i)	Use of Bqv [i.e. $F = 2.15 \times 10^{-14} \text{ N}$] (1) $\frac{mv^2}{r}$ or $m\omega^2 r$ used (1) Radius calculated $r = 3.56 \text{ cm}$ or $F = 3.06 \times 10^{-14} \text{ N}$ or $v = 2 \times 10^7 \text{ m s}^{-1}$ (1) Not circular or radius too large or field not uniform or force not large enough or speed too large (1)			4	4	2	
		(ii)	Starts in correct direction (1) Curving downward or clockwise (1)		2		2		
	(d)		Equation $F = Eq$ used (1) $F = Bqv$ used (1) Note $Eq = Bqv$ or $E = Bv$ scores both marks Substitution of $E = \frac{V}{d}$ (1) (use of $V = Bvd$ gets 3 marks) Final correct calculation(s) e.g. forces = $(2.15 \text{ \& } 2.16) \times 10^{-14} \text{ N}$ or confirmation of V , B , v or d from the other values (1) Final conclusion consistent with answer ecf - true (accept not true because 2.15 different from 2.16) (1)			5	5	3	
			Question 4 total	5	4	9	18	9	1

2022

6. (a) Catrin carries out a Hall effect experiment to find out the number of free electrons per unit volume in a certain metal. She places a wafer of the metal in a known magnetic field and passes a current through it (see diagram).



- (i) By considering the forces acting on free electrons passing through the wafer, explain why a Hall voltage is measured on the voltmeter shown. [4]

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- (ii) By equating the magnetic and electrical forces, show that the Hall voltage, V_H , is given by:

$$V_H = Bvd$$

where B is the magnetic flux density, d is the width of the wafer and v is the drift velocity. [3]

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- (iii) Catrin states that her measured Hall voltage of 68.0 nV is consistent with a drift velocity of approximately 0.07 mms^{-1} . Determine whether, or not, she is correct. [2]

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- (iv) Calculate the number of free electrons per unit volume for the metal. [3]

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- (b) After repeating the same experiment on a different metal, Catrin obtains a value of $(5.85 \pm 0.19) \times 10^{28} \text{ m}^{-3}$, for the number of free electrons per unit volume. She is given a table of values in order to determine which metal has been used in the experiment.

Element	Free electron density / 10^{22} cm^{-3}
Aluminium	18.1
Barium	3.15
Copper	8.47
Gold	5.90
Iron	17.0
Silver	5.86

- Explain which metal(s) she should conclude has been used in the experiment. [2]

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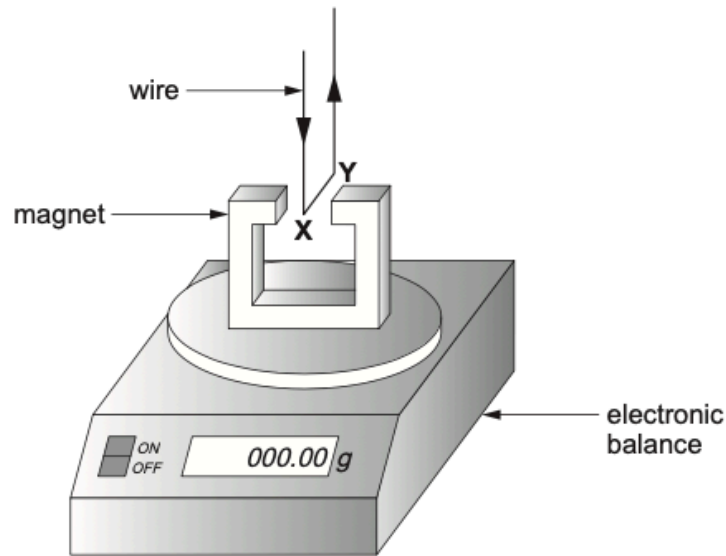
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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	Electrons move at right angles to B -field OR LHR mentioned OR accept Bqv or BIl force (anything to do with magnetic force acting on electrons) (1) Force towards back face or shown clearly in diagram (1) (-ve) charge build (up on back face) (1) Electric field hence voltage (ignore "inducing" = neutral) OR accept equilibrium when mag force = electric force OR accept charge build-up <u>clearly</u> linked to Hall voltage (1)		4		4		4
		(ii)	$Ee = Bqv$ or similar seen e.g. $Eq = BIl$ or conservation of energy stated (1) $E = \frac{V}{d}$ substituted OR $E = \frac{V}{d}$ and $q = It$ and $v = \frac{l}{t}$ substituted or $Vq = Bqvd$ i.e. conservation of energy applied (1) Algebra seen to give correct answer (1)	1	1 1		3	2	
		(iii)	Any 3 numbers substituted to obtain the 4 th (1) Correct answer 4 th number & conclusion seen (1) ($7.26 \times 10^{-5} \text{ m s}^{-1}$ is correct drift velocity) 5.4 mm is calculated d , 65.5 nV is calculated Hall pd, 0.187 T is calculated B -field)			2	2	2	2
		(iv)	Rearrangement of $I = nAve$ ($n = \frac{I}{Ave}$) OR $V_H = \frac{BI}{net}$ known (1) Correct area method i.e. $5.2 \times 0.4 (\times 10^{-6})$ OR $n = \frac{BI}{V_H et}$ (1) Correct answer ($3.28 \times 10^{28} \text{ m}^{-3}$) OR 3.4×10^{28} if they use 0.07 mm s^{-1} (beware ecf may be available for a candidate that uses a previously penalised value instead of the given values)(1)		3		3	3	3

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(b)		1 st mark - could be either <u>gold or silver</u> (both required for the mark) 2 nd mark either: Inside range or reference to uncertainty or range is 5.66 – 6.04 or check the colour of the metal			2	2		2
			Question 6 total	1	9	4	14	7	11

2023

4. The following set-up is used to investigate the force on a current-carrying wire in a magnetic field.



The part **XY** of the wire which is in the magnetic field, has been placed carefully so that the current is from front to back at 90° to the direction of the magnetic field. When there is no current the balance is reset to display 000.00 g. When there is a current in the direction shown, the magnetic force results in a positive reading on the display of the balance. However, this signifies an upward force on the wire.

- (a) (i) Explain briefly why the magnetic force must be upward on the part **XY** of the wire. [2]

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- (ii) **On the diagram**, indicate which is the North pole of the magnet, and state below which rule you used to obtain your answer. [2]

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- (iii) Explain why the vertical parts of the wire have no effect on the reading displayed by the electronic balance. [1]

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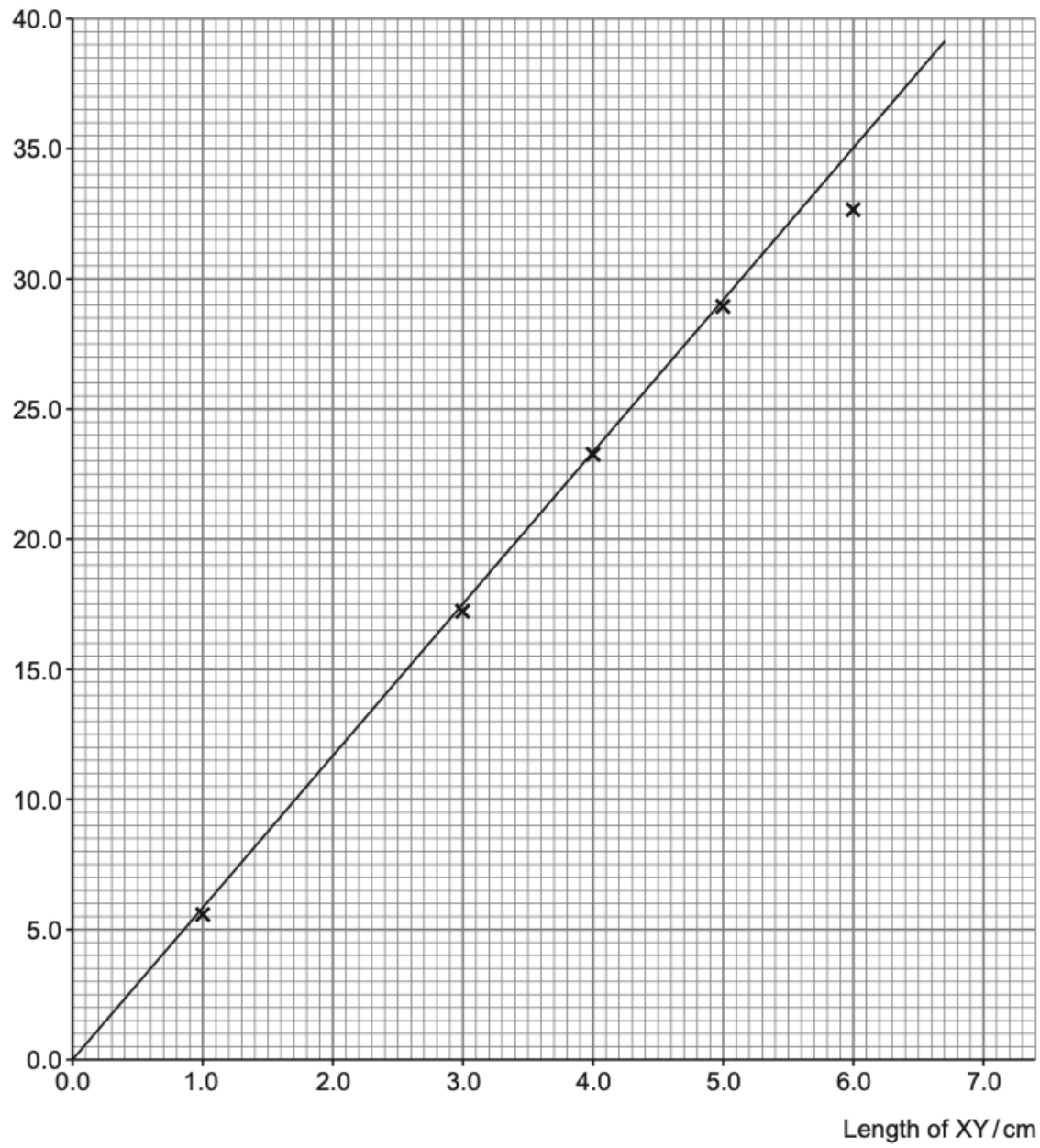
- (b) Bronwen varies the length of XY and records the electronic balance reading each time when a current of 5.00 A flows. She records all her results in a table and plots a graph of magnetic force against length of wire.

(i) **Complete the table and plot** the two missing points on the graph.

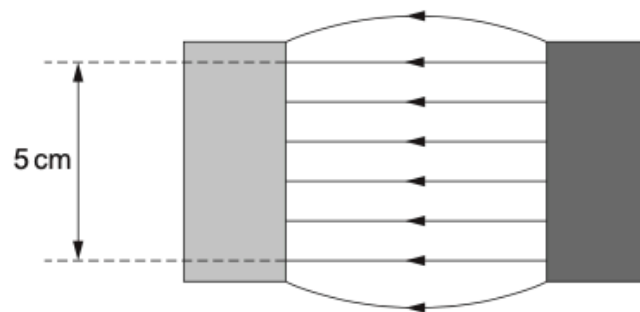
[4]

Length of XY/cm	Balance reading/g	Magnetic force/ 10^{-3} N
1.0	0.56	5.5
2.0	1.22	
3.0	1.75	17.2
4.0	2.36	23.1
5.0	2.89	28.4
6.0	3.32	32.6
7.0		35.8

Magnetic force/ 10^{-3}N



- (ii) The region of uniform magnetic field between the poles of the magnet is of length 5 cm only. Evaluate whether or not Bronwen has drawn the line of best fit correctly. [3]



- (iii) Determine the magnetic flux density, B , of the uniform magnetic field between the poles of the magnet and quote it to an appropriate number of significant figures. The current in the wire is 5.00 A. No uncertainty is required in your value of B . [5]

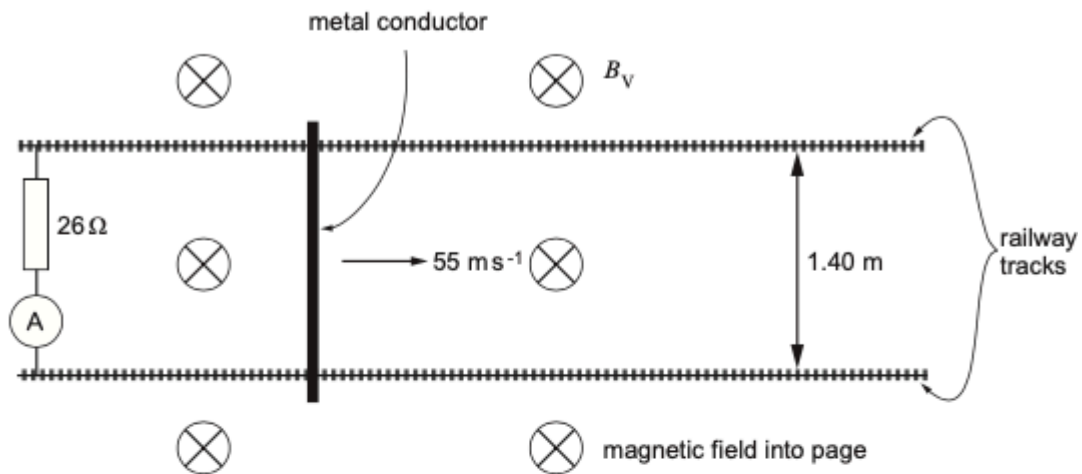
Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)	Force down on magnet or the balance (due to positive) (1) N3 or equal and opposite mentioned (1)		2		2		2
		(ii)	Right side is N (on diagram or stated) (1) [F] LHR (1)	1	1		2		2
		(iii)	Forces are horizontal OR forces are equal and opposite Accept forces are outwards and inwards NOT wire not perpendicular to field		1		1		1
	(b)	(i)	12.0 [$\times 10^{-3}$ N] (1) 3.65 [g] (1) 2 points plotted correctly within half square (2) ecf on 12.0 [$\times 10^{-3}$ N] 1 point plotted correctly within half square (1) 0 points plotted correctly within half square (0)		4		4	4	4
		(ii)	Ignored the last 2 points (1) Since [partly] outside uniform field (1) Award the first 2 marks for: Has only used the points in the uniform field Line good [with same number above and below] (1)			3	3		3

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Conversion of cm (1) Correct method for gradient (1) Gradient = 0.583 (1) Gradient equated to BI (1) Field = 0.115 T unit mark (accept 0.11 to 0.12 to 2sf or 3sf) (1) Note 1.1 T to 1.2 T obtains 4 marks Alternative: Obtaining point on line – don't award for point more than $\frac{1}{2}$ square from line, don't award if point has a length for XY less than 3 cm (1) Converting cm to m (1) Rearranging equation i.e. $B = \frac{F}{Il}$ (1) Substituting (1) Field = 0.115 T unit mark (accept 0.11 to 0.12 to 2sf or 3sf) (1) Note 1.1 T to 1.2 T obtains 4 marks			5	5	5	5
			Question 4 total	1	8	8	17	9	17

Electromagnetic Induction

2017

5. A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component (B_v) of the Earth's magnetic field.



- (a) (i) Explain why a current is detected by the ammeter. [2]

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- (ii) State why the current is independent of the horizontal component of the Earth's magnetic field. [1]

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- (b) Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is $184 \mu\text{A}$. [3]

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(c) State the direction of the current in the resistor and how you obtained this direction. [2]

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(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

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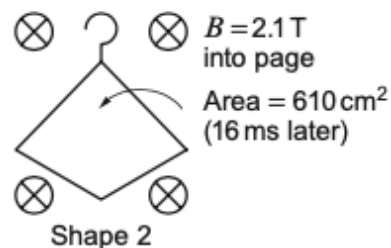
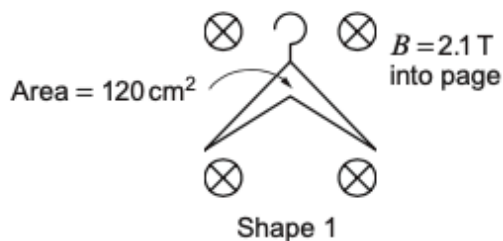
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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)	(i)	Change in flux linkage or cutting flux (1) Complete circuit or implied (1)	2			2		
		(ii)	These are in the plane of motion/the vertical component is perpendicular to motion/these are not cut		1		1		
	(b)		Use of $\frac{BAN}{t}$ or $V = Blv$ (1) Use of $I = \frac{V}{R}$ (1) Answer = $62 \mu\text{T}$ (1)	1 1			3	3	
	(c)		Current down in resistor [or up in conductor or anticlockwise] (1) [F]RHR or [F]LHR (for electrons) or RH grip rule (1)	1	1		2		
	(d)		Other force compared e.g. friction/drag (1) Magnetic force small / resistive force is much bigger (1) Because current small / B small (1) Good calculation e.g. $P = I^2 R = 8.8 \times 10^{-7} \text{ [W]}$ or $F = BIL = 1.6 \times 10^{-8} \text{ [N]}$ ecf on B or calculated values of B using I (1)			4	4	1	
			Question 5 total	5	3	4	12	4	0

2018

5. An experiment is carried out in a very strong uniform magnetic field in order to confirm Faraday's Law under extreme conditions. A coat hanger made of aluminium wire is bent from Shape 1 to Shape 2 in a time of 16 ms.



- (a) (i) Show that a mean emf of 6.4 V is induced in the coat hanger. [2]

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- (ii) Show on the diagram of Shape 2 the direction of the induced current and state very briefly how you determined this direction. [2]

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- (b) The aluminium wire has a circular cross-section of diameter 3.0 mm and the length of the wire through which the current flows is 91 cm. Show that the mean current in the wire is approximately 1900 A (resistivity of aluminium = $2.65 \times 10^{-8} \Omega \text{ m}$). [3]

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- (c) Iestyn claims that changing the shape of the coat hanger from Shape 1 to Shape 2 in 16 ms in the magnetic field will increase its temperature by less than 1 °C. Determine, using appropriate calculations, whether or not Iestyn is correct. (Density of aluminium = 2700 kg m^{-3} , specific heat capacity of aluminium = $897 \text{ J kg}^{-1} \text{ K}^{-1}$.) [5]

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- (d) For medical research, it is decided to investigate the effect of this strong magnetic field (2.1 T) on patients with metal replacement joints to see if the metal joints become hot or undergo large forces (during MRI scans). Discuss the ethics of such an experiment. [3]

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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
5	(a)	(i)	Use of $BA = \text{flux}$ or Faraday's law $\Delta(BA)/t$ (1) [or by impl.] $\frac{2.1 \times (610 - 120) \times 10^{-4}}{0.016}$ seen (or similar) (1)	1	1		2	2	2
		(ii)	Anticlockwise (accept words if not in diagram) (1) FRHR or Right hand screw rule or FLHR on electrons (1) Alternative (for explanation) Lenz's law: [direction of induced flux must oppose so out of paper.]	1	1		2		2
	(b)		Resistance = $\frac{\rho l}{\pi r^2}$ quoted or used (0.0034 Ω) (1) Use of $I = \frac{V}{R}$ (1) $\frac{6.4 \times \pi \times 0.0015^2}{2.65 \times 10^{-8} \times 0.91}$ [=1880 A] or equivalent seen (1)	1	1				
					1		3	3	3
	(c)		$V = \pi r^2 l$ used [= 6.43 $\times 10^{-6} \text{ m}^3$] (1) Mass = density $\times$ volume used [=0.0174 kg] (1) Energy = IVt used (or equivalent 195 J) (1) Energy = $mc\Delta T$ used (1) Answer = 12.5 K so lestyn wrong(1)			5	5	5	5
	(d)		Any 2 $\times$ (1) from - Consent of patient discussed [e.g. volunteers] (1) - possible injury to patient discussed [e.g. long-term effects] (1) - Appropriateness of experiment carried out on model / dead body / animal instead discussed (1) Conclusion Argument is logical and leads to a conclusion (even if a mixed conclusion) (1)			3	3		
			Question 5 total	3	4	8	15	10	12

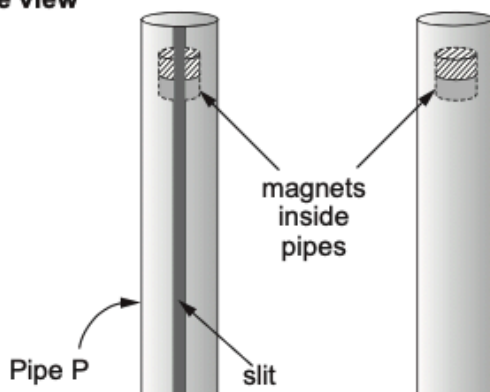
2019

5. (a) State the laws of Faraday and Lenz for electromagnetic induction.

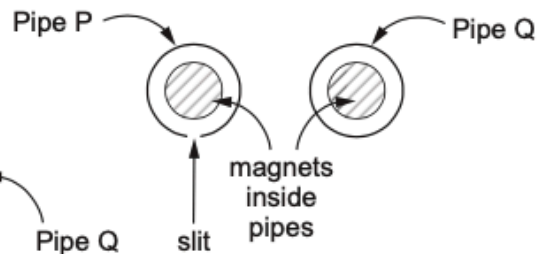
[2]

- (b) Two strong bar magnets are dropped through two copper pipes (P and Q). Pipe P has a slit running along its length but Q is complete. When the magnet is dropped through pipe P it accelerates almost uniformly but the magnet dropped through pipe Q quickly reaches a very low terminal velocity. Explain these observations. [6 QER]

Side view



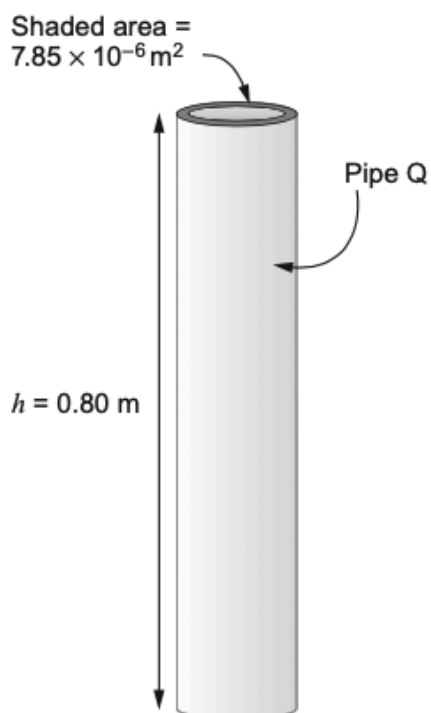
Top view



- (c) By applying the principle of conservation of energy, calculate the temperature increase of pipe Q after the magnet has fallen at constant speed. [4]

Data:

- mass of magnet = 0.300 kg,
- cross-sectional area of the copper walls of pipe Q = $7.85 \times 10^{-6} \text{ m}^2$ (see diagram)
- density of copper = 8960 kg m^{-3}
- specific heat capacity of copper = $385 \text{ J K}^{-1} \text{ kg}^{-1}$.



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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		[Induced] emf (or pd or voltage) equal to (or proportional to) the rate of change (or cutting) of flux [linkage] (1) The emf [tends to] oppose the <u>change</u> [to which it is due] (1)	2			2		
	(b)		Indicative content: Flux in pipe(s) changes or flux cut by pipe(s) Emf induced in pipe(s) Pipe P current cannot flow / incomplete circuit Pipe Q current flows [Current] opposes motion (change)(since) magnetic field set up due to (induced) current Uniform acceleration - no opposing force or gravity only Additional useful points – Terminal velocity - magnetic force equal (and opposite) to weight Slow velocity - due to strong magnet or small resistance or large current GPE converted to internal energy or electrical energy or electromagnetic energy Copper is not magnetic		6		6		

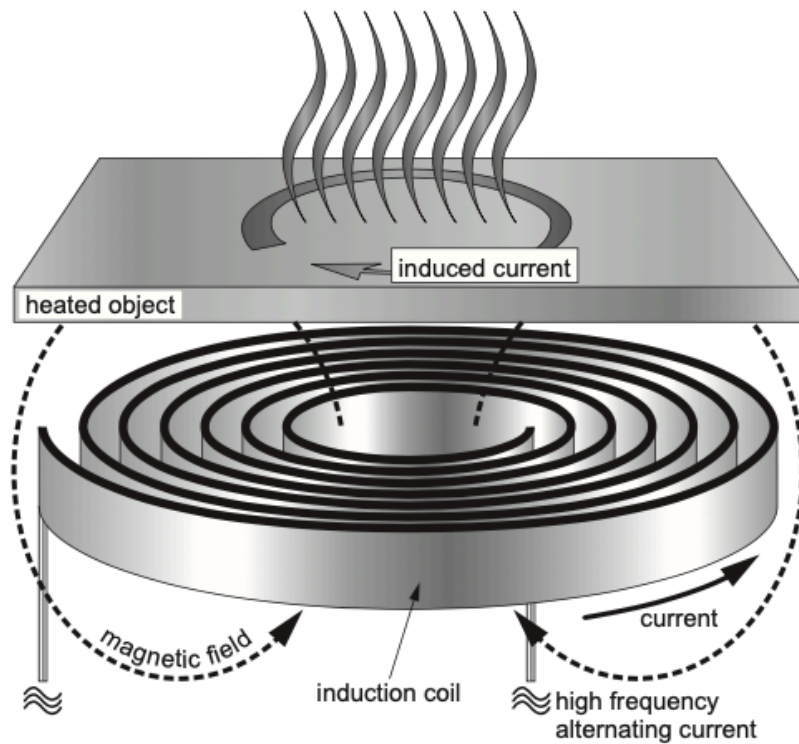
	(c)		mgh or $mc\Delta\theta$ used or implied (i.e. $E_p = 2.35 \text{ J}$) (1) Use of $V = Ah$ (1) Use of $m = \rho V$ (1) (i.e. $m = 0.056 \text{ kg}$) Answer = 0.109 [K] (1)		4		4	4	
			Question 5 total	2	10	0	12	4	0

2022

WJEC 2022 Unit 4 Question 6 (a) (ii)

2023

5. (a) The diagram shows how conductors can be heated using a varying magnetic field.



Explain why the heated object gets hot **and** why a high frequency alternating current is used. [4]

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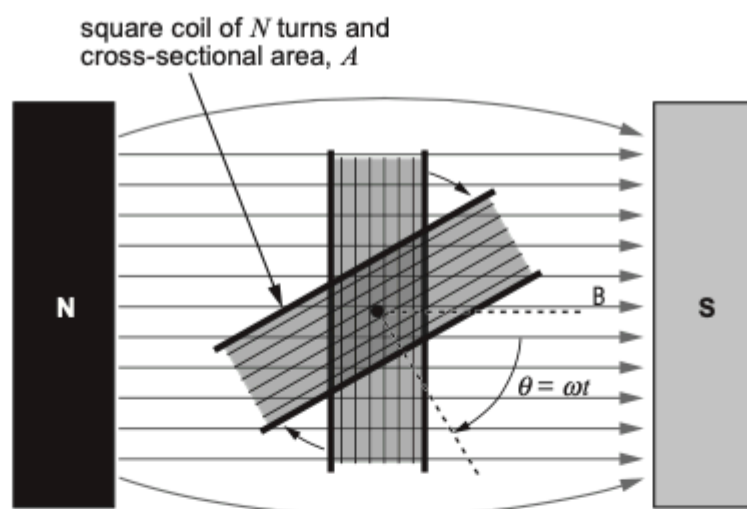
- (b) Explain how a Hall voltage arises in a Hall probe, how it is measured **and** how this can be used to measure the magnetic flux density. A diagram should be included in your answer. [6 QER]

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
5	(a)		Flux in heated object varies (1) Faraday's law referred to or implied (1) Current flows, [providing heating] (1) High frequency increases rate of change of flux (1)	1	1 1 1		4		
	(b)		Indicative content: Diagram Diagram of Hall probe / chip / wafer B -field shown Current flow shown perpendicular Force direction shown correctly Voltmeter correctly connected Slice orientated / adjusted in field Explanation and measuring B Electrons / charge carriers experience force Given by FLHR They move to one side Setting up Hall voltage or field Equilibrium is reached between magnetic and electric forces Hall voltage is proportional to B OR equation quoted Can be calibrated to give B directly OR hence B calculated 5-6 marks Comprehensive diagram and explanation and measuring B . <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i>	6			6		6

Alternating currents [legacy]

2017

6. (a) A coil is rotated in a magnetic field.



- (i) Use Faraday's law to explain why the emf is proportional to the angular velocity of the coil. [2]

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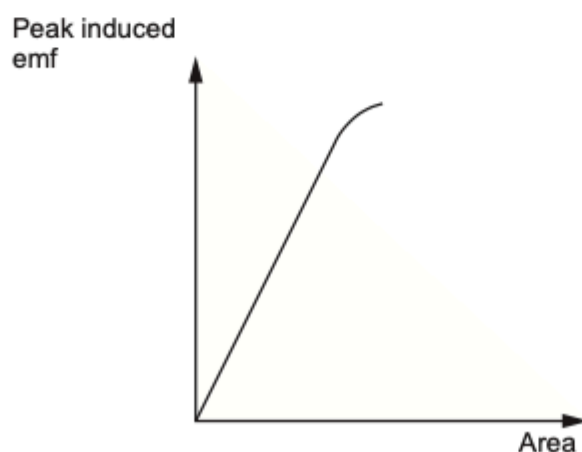
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The peak emf induced in a rotating coil is given by:

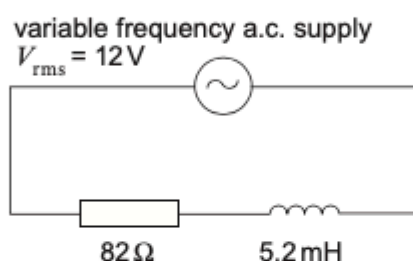
$$E = \omega BAN$$

An experiment is carried out to investigate the variation in the peak emf with area, A , by using coils with different cross-sectional areas. The following graph is obtained.



- (ii) Discuss the agreement of the graph with the equation for the peak emf and suggest a reason for any disagreement (see diagram of coil opposite). [3]

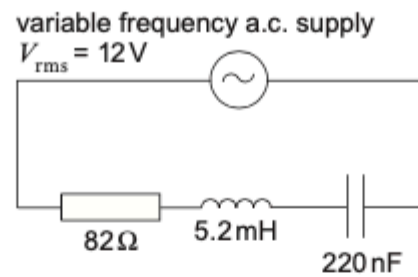
- (b) (i) The minimum impedance of the circuit below occurs when the frequency of the a.c. supply is zero. Explain why this is so. [2]



- (ii) Calculate the rms current when the rms pd across the resistor and inductor are equal. [3]

- (c) (i) Explain how resonance arises in the *LCR* circuit shown.

[3]



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- (ii) Calculate the power dissipated in the circuit at resonance.

[2]

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- (iii) Calculate the rms current when the frequency of the supply is 9.40 kHz.

[3]

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- (iv) Discuss briefly the effectiveness of the given *LCR* circuit for providing a sharp resonance curve. [2]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$.)

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Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)	(i)	Reference to Faraday's law e.g. the emf is the rate of change/cutting of flux/flux linkage (1) The rate of change/cutting of flux (linkage) is proportional to the (angular) velocity (1)	1	1		2		
		(ii)	Agreement since straight line (through origin) or <i>E</i> proportional to <i>A</i> (1) Bad agreement at end (non-linear) (1) Drops off due to non-uniform field or equivalent (1)			3	3	1	
	(b)	(i)	Impedance of the inductor is zero/minimum when $f = 0$ (1) Short justification e.g. only resistance or $Z = R$ or hence $Z = \sqrt{X^2 + 82^2}$ is a minimum (1)		2		2	1	
		(ii)	$Z = \sqrt{82^2 + 82^2} = 116$ (1) $I = \frac{V}{Z}$ (1) Answer $I = 0.103$ [A] (1)	1	1		3	3	
	(c)	(i)	Phasors of capacitor and inductor are opposite direction or capacitor pd in antiphase of inductor (1) Cancel when $X_C = X_L$ or equivalent ($V_C = V_L$) (1) Gives minimum impedance / maximum current (1)	3			3		
		(ii)	Understanding impedance = resistance (1) $P = \frac{V^2}{R} = 1.76$ [W] (1)		2		2	1	
		(iii)	Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2} = 244$ [Ω] (1) Use of $I = \frac{V}{Z}$ (1) Answer = 49 m[A] (1)	1	1		3	3	
		(iv)	Calculation of <i>Q</i> factor (and realisation that this is germane) $Q = 1.9$ (1) Correct evaluation – not a particularly high <i>Q</i> factor or equivalent (can be obtained without <i>Q</i> factor but needs justification e.g. not sharp because resistance a little high (1)			2	2	1	
			Question 6 total	6	9	5	20	10	0

2018

6. (a) The sinusoidal emf produced by a rotating coil in a magnetic field has a **peak** emf of 18.0V and a frequency of 70.0Hz. The coil is connected to the y-input of an oscilloscope.

- (i) The output of the rotating coil is used to power a small lamp of known resistance. Explain how you would calculate the rms input power to the lamp from the known resistance and **peak** emf. [2]

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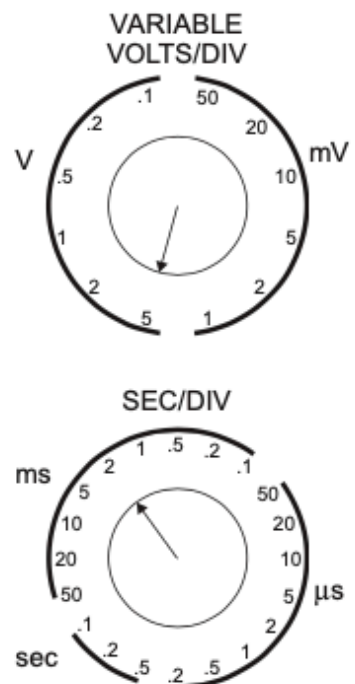
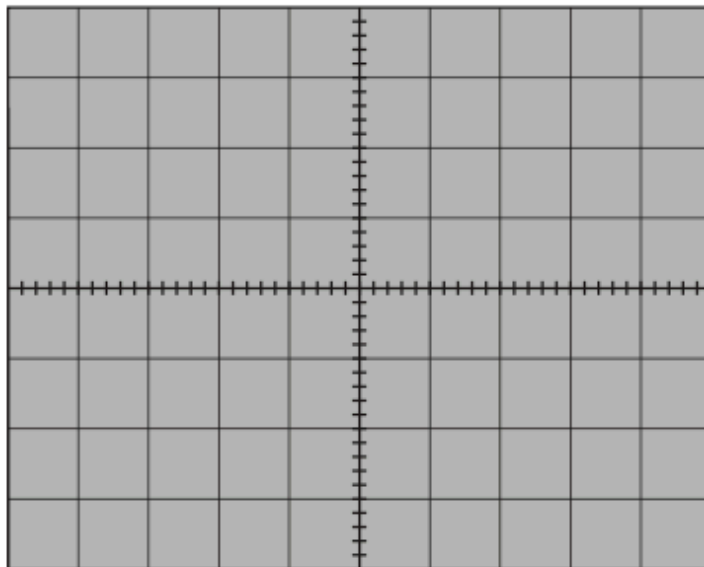
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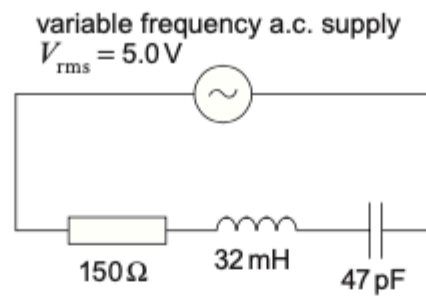
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- (ii) Draw a typical trace that might be seen on the oscilloscope screen with the oscilloscope settings shown. [5]

Space for calculations.



(b) An *LCR* circuit is shown below.



(i) Explain why the resonance frequency of the circuit occurs when: [3]

$$\omega L = \frac{1}{\omega C}$$

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(ii) Calculate the resonance frequency of the circuit. [2]

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(iii) Calculate the rms current when the frequency of the supply is 324 kHz. [3]

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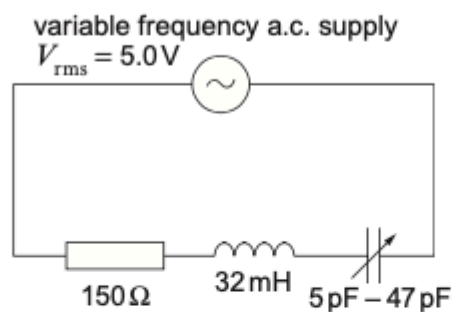
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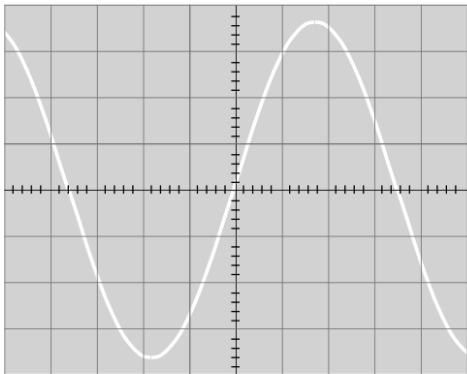
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- (iv) A student claims that the following circuit cannot have a peak pd above 1.5kV across the capacitor. Investigate whether or not he is correct. [5]

(It may be useful to note that $\frac{\omega_0 L}{R} = \frac{1}{R} \sqrt{\frac{L}{C}}$.)

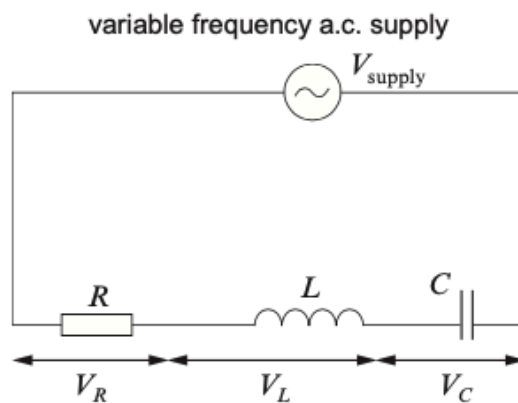


Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
6	(a)	(i)	Rms pd is required or $V_{\text{rms}} = \frac{V}{\sqrt{2}}$ (1) Substitution into $P = \frac{V^2}{R}$ or equivalent e.g. calculating I , $P = IV$ (1)	2			2		
		(ii)	$\frac{18}{5} = 3.6$ squares seen or implied (1) $T = \frac{1}{f}$ used (0.0143 or 14.3 ms) (1) Dividing period by 2 ms (7.1 squares) (1) Period correct in diagram (ecf) (1) accept any phase, at least one cycle Amplitude correct in diagram (ecf) (1) 	1	1 1 1 1		5	2	

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
	(b)	(i)	Reactance of inductor equal and opposite to capacitance (or pds equal and opposite) (1) Minimum impedance (or all pd across resistor) (1) Maximum current (1)	3			3		
		(ii)	Algebra leading to $f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}}$ (1) Answer = 130 kHz (1)		2		2	2	
		(iii)	$X_L = 65\,000\, [\Omega]$ or $X_C = 10\,500\, [\Omega]$ (1) Substitution into $Z = \sqrt{(X_L - X_C)^2 + R^2} = 55\,000\, \Omega$ (1) Use of $I = \frac{V}{Z}$ leading to Answer = 91 μA (ecf) (1)		3		3	3	
		(iv)	Smallest capacitor used (or trial and error) (1) Q factor calculated (533) (or other valid method of getting Q) (1) Pd across C or $L = Q \times 5$ (2 700 V) (1) Peak pd this is $\times \sqrt{2}$ (3 800 V) (1) Student is wrong, it's considerably larger (valid conclusion) (1)			5	5	3	
			Question 6 total	6	9	5	20	10	0

2019

6. (a) The following RCL circuit is constructed.



For this RCL circuit, describe the relationships between the rms pds V_{supply} , V_R , V_L and V_C :

- (i) when the circuit is not in resonance; [3]
Space for diagram.

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- (ii) when the circuit is in resonance. [2]

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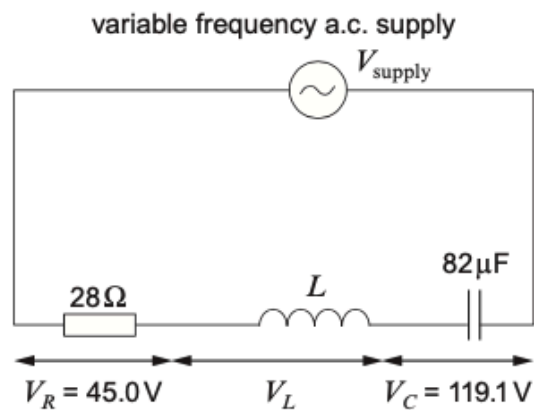
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- (b) The following *RCL* circuit is **at resonance**. The values of *R* and *C* are provided along with their rms pd values.



Calculate:

- (i) the *Q* factor of the circuit; [1]

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- (ii) the rms current; [1]

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- (iii) the frequency of the power supply; [2]

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- (iv) the inductance of the inductor. [2]

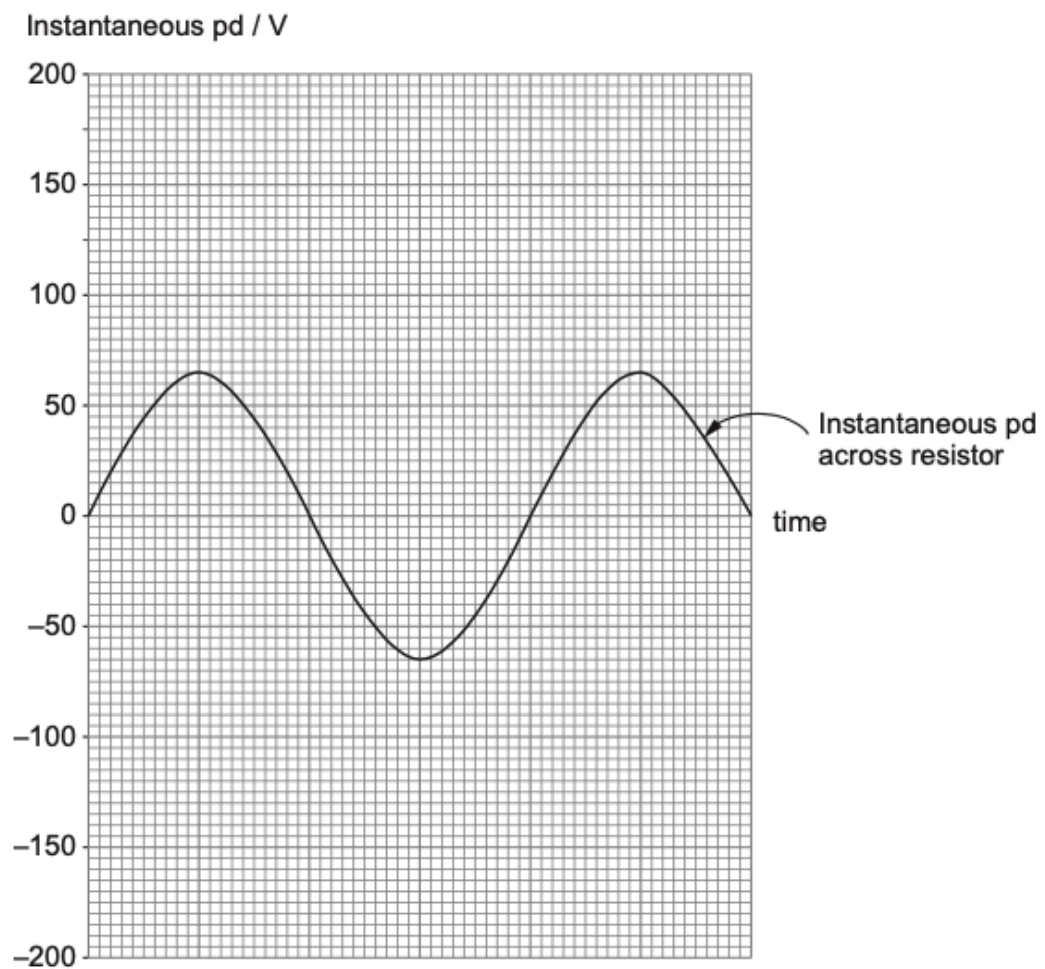
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- (v) Sketch a graph of the instantaneous pd across the capacitor on the grid provided. (The instantaneous pd across the resistor is shown.) [2]



- (vi) Without further calculation, explain why the current decreases when the frequency of the power supply is increased. [2]

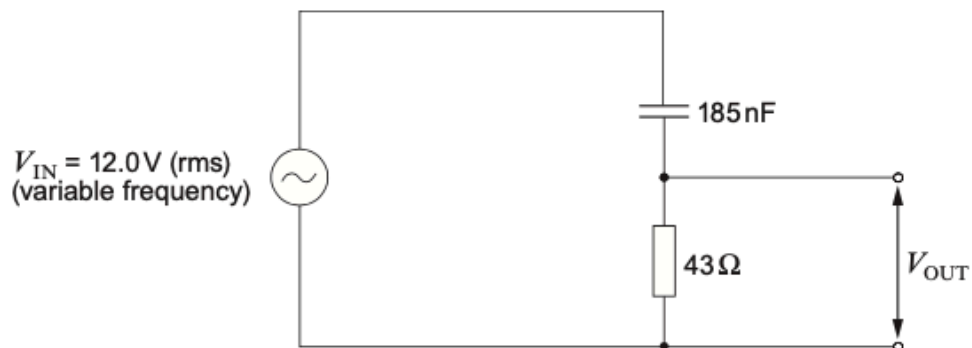
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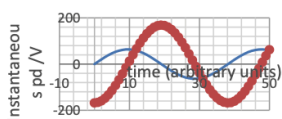
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- (c) Sam claims that the output pd (V_{OUT}) in the following circuit is greater than 6.0V when the frequency is greater than 20 kHz, but less than 6.0V when below 20 kHz. Investigate whether or not Sam is correct. [5]

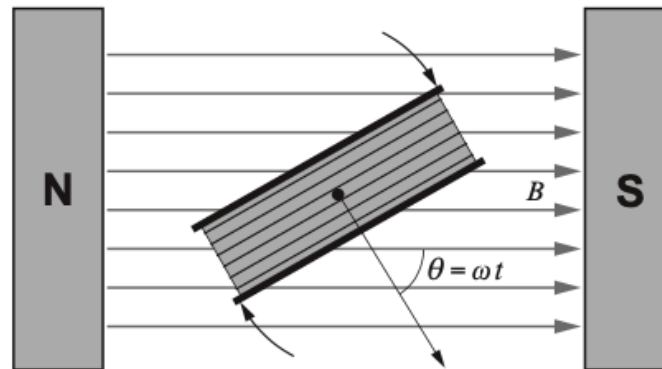


Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
6	(a)	(i)	Pd of L and C are opposite signs (antiphase) (or in phasor diag) (1) Pd of L 90° before $\frac{R}{I}$ and C 90° after $\frac{R}{I}$ (or in phasor diag) (1) Pd of supply is the resultant of all 3 (or in phasor diag) (1)	3			3		
		(ii)	Resonance: R pd is power supply pd (1) Pds of L and C are equal and opposite (1)	2			2		
	(b)	(i)	$\frac{119.1}{45} = 2.65$		1		1	1	
		(ii)	$\frac{45}{28} = 1.61$ [A]		1		1	1	
		(iii)	$X_C = \frac{V}{I}$ used [74.1 Ω] or $X_C = \frac{1}{\omega C}$ (1) Answer = 26.2 [Hz] (1)		2		2	2	
		(iv)	$X_L = \omega L$ used with 74.1 (1) ecf Answer = 0.45 [H] (1)	1	1		2	2	
		(v)	Peak V of 168 V (1) Phase as shown (1) 		2		2	1	
		(vi)	Reactance of inductor increases or reactance of capacitor decreases or $X_L > X_C$ (1) Impedance increases because reactances don't cancel (1)		2		2		

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(c)		Realising frequency variation of capacitor reactance e.g. decreases with frequency or small at high frequency or large at low frequency or explaining $\frac{1}{\omega C}$ (1) Voltage divider argument or equivalent employed leading to low pd at low f and vice versa (1) Calculating reactance at 20 kHz [43.01 Ω] (1) Realising $\sqrt{V_R^2 + V_C^2} = 12$ and not $6\text{ V} + 6\text{ V} = 12\text{ V}$ (1) Correct conclusion Sam is wrong on pds (or should be 8.5 V) but OK on frequency relationship (1)			5	5	3	
			Question 6 total	6	9	5	20	10	0

2023

7. (a) A diagram of a rotating coil in a magnetic field is shown below.



The coil rotates 50 times per second in a field of flux density 0.215 T.
The coil has 2650 turns and cross-sectional area $2.35 \times 10^{-3} \text{ m}^2$. (Note that $\theta = 0$ when $t = 0$)

- (i) Calculate the peak induced emf in the coil. [2]

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- (ii) Calculate the flux linkage of the coil when $t = 5.0 \text{ ms}$.
[Hint: put your calculator in radian mode.] [2]

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- (iii) Calculate the induced emf in the coil when $t = 5.0 \text{ ms}$. [1]

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- (iv) Vanessa states that her answers to parts (ii) and (iii) are consistent with the coil having completed a quarter of a cycle. Determine whether or not she is correct. [3]

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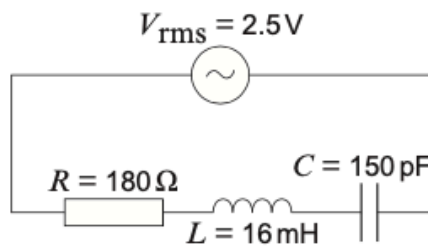
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- (b) (i) Calculate the resonance frequency of the following circuit. [2]



- (ii) Explain why the rms current at resonance is approximately 14 mA. [2]

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- (iii) Calculate the rms current in the circuit when the frequency is 105 kHz. [4]

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- (iv) Explain why your answers to parts (b)(i), (b)(ii) and (b)(iii) suggest that the circuit has a high Q factor. [2]

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- (v) Thomas considers the effect, on this circuit, of increasing the capacitance only. He believes that this will decrease the Q factor. Determine whether or not he is correct. [2]

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Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)	(i)	Choosing correct equation (ωBAN) (1) Correct answer = 421 [V] (1)	1	1		2	1	
		(ii)	Choosing correct equation ($BAN \cos \omega t$) (1) Answer = 0 or close to 0 (1) Award 1 mark for answer of 1.34	1	1		2	1	
		(iii)	421 [V] ecf from (i)		1		1	1	
		(iv)	Period is 20 ms so 5 ms quarter cycle (1) ωt is 90° or equivalent (1) Expect max emf and zero flux (1) Alternative: ωt is 90° or equivalent (1) $\cos 90^\circ = 0$ and $\sin 90^\circ = 1$ (1) Expect max emf and zero flux (1)			3	3		
	(b)	(i)	Choosing correct equation i.e. $\frac{1}{2\pi} \sqrt{\frac{1}{LC}}$ (1) Answer = 103 k[Hz] (1)	1	1		2	1	

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(ii)	Both reactances cancel or total impedance is R or equivalent (1) $I = \frac{V}{Z} = \frac{V}{R}$ (or equivalent) (1)	1	1		2		
		(iii)	Any correct reactance ($X_C = 10\ 105$, $X_L = 10\ 556$) (1) Impedance correctly obtained 485 [Ω] (1) Substitution into $I = \frac{V}{Z}$ ecf on Z (1) Correct answer = 5.15 m[A] ecf on Z (1)	1	1 1		4	4	
		(iv)	Large decrease in current or $\frac{\Delta I}{\Delta f}$ (1) For small increase in frequency or is large (1)	1	1		2	2	
		(v)	Realising ω also changes OR using / deriving $\frac{1}{R} \sqrt{\frac{L}{C}}$ (1) Correctly linking valid equation to obtain yes OR simply yes e.g. in $\frac{\omega L}{R}$ ω decreases so yes (1)			2	2		
			Question 7 total	6	9	5	20	10	0